

Simplicial Complex and Simplicial Homology

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1 Preface

The reason I want to write this note is that the books introducing simplicial complex are not very suitable for my study. Some books omit too many details and are too ambiguous (e.g. Allen Hatcher's Algebraic Topology. To be honest Hatcher used Δ -complex to talk about simplicial complex, which is not very common), while others are redundant (e.g. Munkres' Elements of Algebraic Topology). Spanier's Algebraic Topology is a very good reference - it is very rigorous and its perspective is at a very high level, but it also omits some of the steps of proofs and detailed explanations which will make the book very elusive - it looks more like a dictionary though. That is why I want to summarize my own version. This note is based on Prof. Chi Chen-yu's lectures as well as Chapter 3 of Spanier's Algebraic Topology, but provides more details about the proof. This note is not guaranteed to be all correct, so please email me if you find anything incorrect.

2 Weak Topology

First of all, we need to introduce the concept of weak topology before diving into Chapter 3.1 of Spanier.

Definition 2.1. Let X be a set. $\{A_\alpha | \alpha \in \mathcal{A}\}$ is a family of subsets of X , each of which has a specific topology and an injection map $i_\alpha : A_\alpha \rightarrow X$. The weak topology on X induced by $\{A_\alpha | \alpha \in \mathcal{A}\}$ is defined as follows: $U \subset X$ is open $\iff U \cap A_\alpha$ is open for all $\alpha \in \mathcal{A}$.

Remark 2.1. It is very trivial to prove that weak topology satisfies the definition of topology.

Remark 2.2. Equivalently, C is closed in $X \iff C \cap A_\alpha$ is closed in A_α for all $\alpha \in \mathcal{A}$. The proof is trivial.

Remark 2.3. This is the finest topology on X such that i_α is continuous for all $\alpha \in \mathcal{A}$.

Example 2.1. Let $\{A_\alpha | \alpha \in \mathcal{A}\}$ be a family of topological spaces. The topology of the disjoint union $\bigsqcup_{\alpha \in \mathcal{A}} A_\alpha$ is defined as the weak topology.

Proposition 2.1. X is a topological space with weak topology induced by $\{A_\alpha | \alpha \in \mathcal{A}\}$. For any topological space Y , a map $f : X \rightarrow Y$ is continuous $\iff f \circ i_\alpha : A_\alpha \rightarrow Y$ is continuous for all $\alpha \in \mathcal{A}$.

Proof. $\Rightarrow$: The composition of two continuous maps is continuous.

$\Leftarrow$: Let U be open in Y . Then $i_\alpha^{-1}(f^{-1}(U))$ is open in A_α for all $\alpha \in \mathcal{A} \iff f^{-1}(U)$ is open in X . $\square$

Proposition 2.1 provides a good way to verify if a map is continuous from X with weak topology to Y a topological space. If we apply this proposition to the disjoint union example, we will immediately get the disjoint union is the coproduct in the category of topological spaces. We will use this proposition many times as we proceed.

Example 2.2. $A_\alpha \xrightarrow{i_\alpha} \bigsqcup_{\alpha \in \mathcal{A}} A_\alpha$ is the coproduct in the category of topological spaces.

Proof. Use Proposition 2.1 and the conclusion is immediate. $\square$

Proposition 2.2. For $\alpha, \beta \in \mathcal{A}$, let $A_{\alpha\beta}$ denote $A_\alpha \cap A_\beta$ equipped with the subspace topology induced by A_α . If for all $\alpha, \beta \in \mathcal{A}$, $A_{\alpha\beta}$ is a closed subset of A_α , $A_{\beta\alpha}$ is a closed subset of A_β and the identity map $id : A_{\alpha\beta} \rightarrow A_{\beta\alpha}$ is a homeomorphism, then A_α is closed in X with respect to weak topology for all $\alpha \in \mathcal{A}$. The induced subspace topology on A_α from weak topology coincides the original topology.

This proposition is very important. It gives a sufficient condition in which the subspace topology is equal to the original topology on each A_α . If you regard A_α as the building block to construct the weak topology on X , then the subspace topology of weak topology restricted on each building block does not change if the condition in proposition 2.2 holds. It also tells you that if this condition holds each building block is closed under weak topology.

Proof. Use remark 2.2: to proof that A_α is closed under weak topology for each $\alpha \in \mathcal{A}$, we only need to verify that $A_\alpha \cap A_\beta$ is closed in A_β for all $\beta \in \mathcal{A}$, and this is exactly what has been stated in the proposition.

The second half of the proof is to prove the subspace topology on A_α is identical with the original topology for each $\alpha \in \mathcal{A}$. We only need to prove that they have the same closed sets. Suppose U is closed in A_α under subspace topology. Then $U = W \cap A_\alpha$ where W is closed in X . By remark 2.2, W is closed in $X \iff W \cap A_\beta$ is closed in A_β for all $\beta \in \mathcal{A}$. Therefore, $U = W \cap A_\alpha$ is closed in A_α .

Conversely, suppose U is closed in A_α . If we can prove $U \cap A_\beta$ is closed in A_β for all $\beta \in \mathcal{A}$, then we can conclude that U is closed in weak topology of X . Since $U \in A_\alpha$, U is closed in subspace topology. Since $U = U \cap A_\alpha$ is already closed in A_α , we assume that $\beta \neq \alpha$. $U \cap A_\beta = (U \cap A_\alpha) \cap A_\beta = U \cap (A_\alpha \cap A_\beta) = U \cap A_{\alpha\beta}$. Since U is closed in A_α , then $U \cap A_{\alpha\beta}$ is closed in $A_{\alpha\beta}$. We also know that the identity map between $A_{\alpha\beta}$ and $A_{\beta\alpha}$ is a homeomorphism, so $U \cap A_{\alpha\beta}$ is also closed in $A_{\beta\alpha}$. Since $A_{\beta\alpha}$ is closed in A_β , then $U \cap A_{\alpha\beta} = U \cap A_\beta$ is closed in A_β , so U is closed under subspace topology. $\square$

3 Simplicial Complex

There isn't a lot of additional things to say about this section before the realization of simplicial complexes. Therefore I will follow Chapter 3 of Spanier's Algebraic Topology until standard realization of simplicial complexes. The way of defining simplicial complex is exactly the same as Spanier, but in a clearer way.

Definition 3.1. A **simplicial complex** K consists of

- (1). a set V_k which is called the vertices of K .
- (2). a set K of **finite nonempty** subsets of V_k such that
 - (2.1). $\{v\} \in K$ for all $v \in V_k$
 - (2.2). If s' is a nonempty subset of $s \in K$, then $s' \in K$

We call $s \in K$ a q -simplex of K if $\#s = q + 1$. $\forall \emptyset \neq s' \subset s \in K, s' = s \iff \dim s' = \dim s$. Now let us introduce some examples. These examples are all listed in Spanier as well.

Example 3.1. $V_k = \emptyset, K = \emptyset$

Example 3.2. Let A be a set. $V_k = A$ and $K = \{s \subset A | 0 < \#s < \infty\}$

Example 3.3. Let K be a simplicial complex and $s \in K$

- (a). $\bar{s} = \{s' | \emptyset \neq s' \subset s\}$
- (b). $\overset{\circ}{s} = \{s' | \emptyset \neq s' \subsetneq s\}$
- (c). $K^q = \{s \in K | \#s \leq q + 1\}$. This is called the q dimensional skeleton of K .

Example 3.4. Let K_1 and K_2 be simplicial complexes. The **join** of K_1 and K_2 , $K_1 * K_2 = (K_1 \sqcup K_2) \cup \{s_1 \sqcup s_2 | s_1 \in K_1, s_2 \in K_2\}$.

These examples are useful in the following proofs so they are listed here. There are many other examples such as nerves which are used to define Cech homology. They are not important and will not be used here so please check out Spanier for more examples. Here I will define the most useful terms.

Definition 3.2. Let K be a simplicial complex. $\dim K = \begin{cases} -1, & K = \emptyset \\ \sup\{\dim s | s \in K\}, & \text{otherwise} \end{cases}$.

We say that K is **finite** if $\#K < \infty$ ($\#K < \infty \Rightarrow \dim K < \infty$).

Definition 3.3. Let K_1 and K_2 are two simplicial complexes. V_{K_1} and V_{K_2} are the vertices of each simplicial complexes respectively. A map $\varphi : V_{K_1} \rightarrow V_{K_2}$ is called a **simplicial map** from K_1 to K_2 if $\forall s \in K_1, \varphi(s) \in K_2$.

Definition 3.4. The **category of simplicial complexes (SC)** consists of:

- (a). objects: $\text{Ob}(SC)$, all simplicial complexes, and
- (b). morphisms: $\text{Hom}_{SC}(K_1, K_2)$, all simplicial maps from K_1 to K_2 .

Please verify that it is a category. It should be obvious.

Definition 3.5. Let K be a simplicial complex and $L \subset K$. We say that L is a **subcomplex** of K if for any nonempty subset s' of $s \in L$, $s' \in L$.

Example 3.5. K is a simplicial complex, $s \in K$, q is a non-negative integer. Then $\bar{s}$, $\overset{\circ}{s}$ and K^q are all subcomplexes of K .

Example 3.6. $\{L_\alpha\}_{\alpha \in A}$ is a family of subcomplexes of a simplicial complex K . Then $\cap_{\alpha \in A} L_\alpha$ and $\cup_{\alpha \in A} L_\alpha$ are both subcomplexes of K .

In this section, I list several definitions above. They seem to be very boring because I did not add any explanation here. Some readers may ask why we need to define simplicial complex like this? To be honest, the definition here is the most rigorous one which is also refereed to as "abstract simplicial complex" in some literature (e.g. An introduction to algebraic topology by Rotman and Elements of algebraic topology by Munkres). They also introduce the concept in a more intuitive way (embedding in Euclidean space). In fact, this intuitive way of introduction is widely used in engineering (meshing technology). If you like a more intuitive definition, please refer to their books.

4 The Standard Realization of Simplicial Complexes

So far a simplicial complex and a topological space has nothing to do with each other. In this section I will introduce the realization of simplicial complexes which is a set, and define a (weak) topology on it. You may know the concept of "mesh generation" if you study mechanical engineering (a step in finite element method). It is basically a method to partition a bounded region. The concept is a generalization of it. For more details, check Wikipedia.

Definition 4.1. *Let K be a simplicial complex. The **standard realization** of K , denoted as $|K|$, is*

$$|K| = \begin{cases} \emptyset, & K = \emptyset \\ \{\alpha : V_k \rightarrow [0, 1] \mid \sum_{v \in V_k} \alpha(v) = 1 \text{ and } \text{supp}(\alpha) \in K\}, & K \neq \emptyset \end{cases}$$

*That is, for any nonempty K , the realization $|K|$ is a set of functions from V_k to closed interval $[0, 1]$ such that for each function $\alpha \in |K|$, $\sum_{v \in V_k} \alpha(v) = 1$ and the **support** of α , $\text{supp}(\alpha) = \{v \in V_k \mid \alpha(v) \neq 0\}$ is in K .*

Remark 4.1. *Note that for any $\alpha : V_k \rightarrow [0, 1]$, if α satisfies the condition $\text{supp}(\alpha) \in K$, there are only finite $v \in V_k$ such that $\alpha(v) \neq 0$. This is because each element of K is a finite subset of V_k (review definition 3.1 if you don't remember this). In this case, $\sum_{v \in V_k} \alpha(v) = 1$ is a finite sum.*

The definition of $|K|$ is extremely abstract and will confuse many people. Now I will give a clearer way to illustrate this definition. In fact, we can associate every $v \in V_k$ with the corresponding characteristic function $\chi_{\{v\}}$. The characteristic function $\chi_{\{v\}}$ is a function from V_k to $[0, 1]$ such that $\chi_{\{v\}}(v') = 0$ if $v \neq v'$ and $\chi_{\{v\}}(v) = 1$. Obviously, $\chi_{\{v\}} \in |K|$ for all $v \in V_k$. Now we can use these characteristic functions to express every $\alpha \in |K|$: $\alpha = \sum_{v \in V_k} \alpha(v) \chi_{\{v\}}$. With this tool, I will show some simple examples here.

Example 4.1. *Consider the following simplicial complex:*

$$V_k = \{A, B, C\}, \quad K = \{\{A\}, \{B\}, \{C\}, \{A, B\}, \{A, C\}, \{B, C\}, \{A, B, C\}\}.$$

Let us see what does $|K|$ look like. Suppose $\alpha : V_k \rightarrow \mathbb{R}$. We can express α as $\alpha = \alpha(A) \chi_{\{A\}} + \alpha(B) \chi_{\{B\}} + \alpha(C) \chi_{\{C\}}$. Obviously, the set of functions from V_k to $\mathbb{R}$ forms a 3 dimensional vector space with a basis $\{\chi_{\{A\}}, \chi_{\{B\}}, \chi_{\{C\}}\}$. If we add the condition $\sum_{v \in V_k} \alpha(v) = \alpha(A) + \alpha(B) + \alpha(C) = 1$, then the set of functions satisfying this constraint exactly forms the

triangle (with interior) with vertex $\chi_{\{A\}}, \chi_{\{B\}}$ and $\chi_{\{C\}}$ in this 3 dimensional vector space. Therefore, if we add another constraint $\text{supp}(\alpha) \in K$, then the set of functions satisfying both constraints should be the subset of this triangle. One could easily verify that all the points in this triangle satisfy the condition $\text{supp}(\alpha) \in K$. Therefore, $|K|$ can be described as a triangle (with interior) whose vertices are $\chi_{\{A\}}, \chi_{\{B\}}$ and $\chi_{\{C\}}$.

Example 4.2. Similarly, one can also verify that if $V_k = \{A, B, C\}$, $K = \{\{A\}, \{B\}, \{C\}, \{A, B\}, \{A, C\}\}$, then $|K|$ is formed by the union of two line segments: the line segment connected by $\chi_{\{A\}}$ and $\chi_{\{B\}}$ and the line segment connected by $\chi_{\{B\}}$ and $\chi_{\{C\}}$ in the 3 dimensional vector space described in Example 4.1.

By the two examples above, one may have realized that the realization of a simplicial complex is the "union of simplices", where a "0 dimensional simplex" is a point, "1 dimensional simplex" is a line segment, "2 dimensional simplex" is a triangle, "3 dimensional simplex" is a tetrahedron and so on. This is the generalization of "polyhedra" (I haven't define this term yet, but you can imagine by common sense). One may ask a natural question: if L is a subcomplex of K , L should also have a realization $|L|$. Can $|L|$ be viewed as a subset of $|K|$? The answer is yes.

Proposition 4.1. If $L \subset K$ is a subcomplex, there exists a canonical injection $i_{KL} : |L| \hookrightarrow |K|$,

$$i_{KL}(\alpha)(v) = \begin{cases} \alpha(v), & v \in V_L, \\ 0, & v \notin V_L. \end{cases}$$

Apparently, $\sum_{v \in V_K} i_{KL}(\alpha)(v) = \sum_{v \in L} \alpha(v) = 1$ and $\text{supp}(i_{KL}(\alpha)) = \text{supp}(\alpha) \in L \subset K$. That is why the image of i_{KL} lies in $|K|$. It is also obvious that i_{KL} is injective. Therefore, $|L|$ can be viewed as a subspace of $|K|$. There is an equivalent description of the $|L|$ by the following proposition.

Proposition 4.2. $i_{KL}(|L|) = \{\alpha \in |K| \mid \text{supp}(\alpha) \in L\}$

Proof. Left to right is immediate by definition of i_{KL} . Suppose $\text{supp}(\alpha) \in L$, then the restriction of α on V_L lies in $|L|$, and i_{KL} brings it to α again, so $\alpha \in i_{KL}(|L|)$. $\square$

Now, since $|K|$ is a set, we would like to make it a topological space. In fact, we can define two topologies on it.

Definition 4.2. Let $d : |K| \times |K| \rightarrow \mathbb{R}$, $d(\alpha, \beta) = \sum_{v \in V_k} |\alpha(v) - \beta(v)|$. Obviously d makes $|K|$ a metric space with induced topology. We denote this topological space $|K|_d$.

Before jumping into the second topology of $|K|$, let us define some notation. Let $s \in K$. $|s| := \{\alpha \in |K| \mid \text{supp}(\alpha) \subset s\}$, $\langle s \rangle := \{\alpha \in |K| \mid \text{supp}(\alpha) = s\}$. Note that the first is the realization of the subcomplex $\bar{s}$. Suppose $s = \{v_0, v_1, \dots, v_q\}$, we can actually explicitly express $|s|$ and $\langle s \rangle$ by the following lemma.

Lemma 4.1. Let $s = \{v_0, v_1, \dots, v_q\}$, then $|s| = \{\alpha : V_k \rightarrow [0, 1] \mid \alpha = \sum_{i=0}^q t_i \chi_{\{v_i\}}, \sum_{i=0}^q t_i = 1, 0 \leq t_i \leq 1, \forall i \in \{0, 1, \dots, q\}\}$. If $q > 0$, then $\langle s \rangle = \{\alpha : V_k \rightarrow [0, 1] \mid \alpha = \sum_{i=0}^q t_i \chi_{\{v_i\}}, \sum_{i=0}^q t_i = 1, 0 < t_i < 1, \forall i \in \{0, 1, \dots, q\}\}$. In other words, $|s|$ and the standard q -simplex Δ^q have a one-to-one correspondence; $\langle s \rangle$ and the interior of Δ^q have a one-to-one correspondence.

Proof. Let us start with $|s|$. For any $\alpha \in |s|$, $\alpha = \sum_{v \in V_k} \alpha(v) \chi_{\{v\}}$ subjected to $\sum_{v \in V_k} \alpha(v) = 1$ and $0 \leq \alpha(v) \leq 1, \forall v \in V_k$. Due to the condition $\text{supp}(\alpha) \subset s$, we can rewrite the expression as $\alpha = \sum_{i=0}^q \alpha(v_i) \chi_{\{v_i\}}$ subjected to $\sum_{i=0}^q \alpha(v_i) = 1$ and $0 \leq \alpha(v_i) \leq 1, \forall i \in \{0, 1, \dots, q\}$. Denote $t_i = \alpha(v_i)$ and we get $\alpha \in \{\alpha : V_k \rightarrow [0, 1] | \alpha = \sum_{i=0}^q t_i \chi_{\{v_i\}}, \sum_{i=0}^q t_i = 1, 0 \leq t_i \leq 1, \forall i \in \{0, 1, \dots, q\}\}$. Conversely, if $\alpha \in \{\alpha : V_k \rightarrow [0, 1] | \alpha = \sum_{i=0}^q t_i \chi_{\{v_i\}}, \sum_{i=0}^q t_i = 1, 0 \leq t_i \leq 1, \forall i \in \{0, 1, \dots, q\}\}$, then $\alpha(v) = t_i$ if $v \in s$ and $\alpha(v) = 0$ if $v \notin s$. This means $\text{supp}(\alpha) \subset s$, so $\alpha \in |s|$.

The proof is similar for $\langle s \rangle$. The only difference from the proof above is: when we proof left is included by right, $\text{supp}(\alpha) = s$ implies $\alpha(v_i) > 0, \forall v_i \in s$. On the other hand $\alpha(v_i) < 1$ because otherwise $\alpha(v_i) = 0$ for some vertex $v_i \in s$ due to the restriction $\sum_{i=0}^q \alpha(v_i) = 1$. The other direction is similar. $\square$

$|s|$ is a subset of $|K|_d$. Therefore, the restriction of the metric on $|s|$ makes $|s|$ a metric space as well. This metric space is denoted $|s|_d$. To be specific, let us choose two points α and β from $|s|_d$ and their distance is $d(\alpha, \beta) = \sum_{i=0}^q |\alpha(v_i) - \beta(v_i)|$. Therefore, $|s|_d$ is homeomorphic (isometric) to the standard q -simplex Δ^q with standard Euclidean topology induced by 1-norm, so $|s|_d$ is compact. There is one thing you should be aware of: Spanier uses "2-norm" instead of "1-norm" to define d but they should induce the same topology by norm equivalence theorem in functional analysis. Since $|s|_d$ are subsets of $|K|$, they induces a weak topology on $|K|$. That is, U is closed in $|K|$ if and only if $U \cap |s|$ is closed in $|s|_d$ for all $s \in K$. By the property of weak topology, the inclusion map $i_s : |s| \rightarrow |K|$ is continuous and the weak topology is the finest topology on $|K|$ such that all injections i_s are continuous.

Let us apply proposition 2.2 to determine the relationship between $|s|$ with metric topology and subspace topology induced by weak topology: if $s_1 \in K$ and $s_2 \in K$, then $|s_1 \cap s_2| = \{\alpha \in |K| | \text{supp}(\alpha) \subset s_1 \cap s_2\} = \{\alpha \in |K| | \text{supp}(\alpha) \subset s_1\} \cap \{\alpha \in |K| | \text{supp}(\alpha) \subset s_2\} = |s_1| \cap |s_2|$ equipped with the same subspace topology induced from $|s_1|_d$ and $|s_2|_d$, and $|s_1 \cap s_2|$ is closed in $|s_1|_d$ (and $|s_2|_d$ respectively). Therefore, the subspace topology on $|s|$ induced by $|K|$ is exactly the same as $|s|_d$.

Let $|K|$ denote $|K|$ equipped with the weak topology induced by $\{|s|_d | s \in K\}$. (Yeah this sounds weird but is not wrong. Typically it is better to use $|K|_w$ to denote the weak topology but Prof. Chi Chen-yu didn't use this notation. I will keep it the same with Prof. Chi) Let us study the weak topology of $|K|$.

Lemma 4.2. $|s|$ is closed in $|K|_d$.

Proof. Consider the natural inclusion $i : |s|_d \rightarrow |K|_d$. $|K|_d$ is a metric space, so it is Hausdorff. $|s|_d$ is compact. A continuous map from compact set to Hausdorff is a closed map, so $|s|$ is closed in $|K|_d$. $\square$

Remark 4.2. $|s|$ is called closed simplex in $|K|$ because it is closed in $|K|$. $\langle s \rangle$ is called open simplex but it is not necessarily an open set in $|K|$!

Corollary 4.1. The identity map $I(K) : |K| \rightarrow |K|_d$ is continuous.

Proof. Use proposition 2.1: The identity map $I(K)$ is continuous iff $i(s) : |s|_d \rightarrow |K|_d$ is continuous. $\square$

Corollary 4.2. $|K|$ is Hausdorff.

Proof. Let α and β be two distinct points in $|K|$. We know that $|K|_d$ is Hausdorff, so there exists two non-intersected open neighborhoods U_α and U_β of α and β respectively in $|K|_d$. Due to the continuity of $I(K)$, U_α and U_β are two non-intersected open sets in $|K|$. $\square$

Lemma 4.3. *Let L be a subcomplex of K . Then $|L|$ is closed in $|K|_d$.*

Proof. To prove $|L|$ is closed in $|K|_d$, we need to prove $|K| - |L|$ is open in $|K|_d$. Let $\alpha \in |K| - |L|$, so $\text{supp}(\alpha) \notin L$. Let $m := \min \text{supp}(\alpha) > 0$. Then for all $\beta \in |K|_d$ and $d(\alpha, \beta) < \frac{m}{2}$, $\text{supp}(\alpha) \subset \text{supp}(\beta)$. Thus, $\text{supp}(\beta) \notin L$ and hence $\beta \in |K| - |L|$. This means there is a open neighborhood of α in $|K| - |L|$. Therefore, $|K| - |L|$ is open in $|K|_d$. $\square$

Corollary 4.3. *$|L|$ is closed in $|K|$.*

Proof. We already proved that $|L|$ is closed in $|K|_d$. Since the identity map $I(K)$ is continuous, so $|L| = I(K)^{-1}(|L|)$ is closed in $|K|$. $\square$

There is one thing to be aware of: the subspace topology on $|L|$ induced by weak topology is generally not equivalent to the weak topology generated by $\{|s|_d | s \in L\}$. Therefore we need to be clear which topology we use on $|L|$. In general, the topology we consider is the subspace topology on $|L|$ by default unless specified, e.g., the following corollary.

Corollary 4.4. *Let Y be a topological space and $f : |K| \rightarrow Y$ be a map. Then f is continuous $\iff f|_{|s|} : |s|_d \rightarrow Y$ is continuous for all $s \in K \iff f|_{|K|^q} : |K|^q \rightarrow Y$ is continuous for all $q \in \mathbb{N} \cup \{0\}$. Here the topology of $|K|^q$ is weak topology induced by $\{|s|_d | s \in K^q\}$.*

The next theorem is very important as it states the structure of $|K|$.

Theorem 4.1. $|K| = \sqcup_{s \in K} \langle s \rangle$.

Proof. First, let us prove that if $\langle s_1 \rangle \cap \langle s_2 \rangle = \emptyset$ if $s_1 \neq s_2$. This is obvious since if $\text{supp}(\alpha)$ cannot be simultaneously equal to s_1 and s_2 . The equivalence is also immediate by definition. $\square$

Likewise, we have the same theorem for subcomplex.

Theorem 4.2. *If L is a subcomplex of K , then $|L| = \sqcup_{s \in L} \langle s \rangle$.*

Definition 4.3. *Let $\alpha \in |K|$ and $s := \text{supp}(\alpha) \in K$. Therefore $\alpha \in \langle s \rangle$. We call $\langle s \rangle$ to be the **carrier** of α . Every element $\alpha \in |K|$ has a unique carrier according to Theorem 4.1.*

In my opinion, the lemmas and propositions are difficult for the readers if this is the first time studying (abstract) simplicial complex. However, it becomes trivial if you are more familiar with simplicial complex. In fact, you should be able to prove all these lemmas and propositions on your own if you are familiar with it enough. Next, we will prove a very important proposition for simplicial complex. What does a compact subset in $|K|$ look like?

Lemma 4.4. *For any subset $A \subset |K|$, $\exists B \subset A$, s.t. B is closed and discrete (i.e. every subset in B is open (and closed)) in $|K|$ and*

$$\#(B \cap \langle s \rangle) = \begin{cases} 1, & \text{if } A \cap \langle s \rangle \neq \emptyset, \\ 0, & \text{if } A \cap \langle s \rangle = \emptyset. \end{cases}$$

Proof. B can be easily constructed. We can choose a point in $A \cap \langle s \rangle$ for each $s \in K$ if $A \cap \langle s \rangle$ is not empty, and let the set to be B . B is closed because $B \cap |s|$ is finite (because $|s| = \sqcup_{s' \in s} \langle s' \rangle$ and the union is finite). For each subset of B , it is also closed in $|K|$ (proof is the same as B) so B is discrete. $\square$

Corollary 4.5. *If A is a compact set in $|K|$, then A must intersect with finite number of open simplices in $|K|$.*

Proof. If A is compact, then all its closed subset should also be compact. Since B is closed in $|K|$, B is closed in A and hence B is compact discrete set which should be finite. $\square$

Using this corollary, we can immediately conclude the following two results describing the compactness of (sub)complex.

Lemma 4.5. *L is a subcomplex of $|K|$. $|L|$ is compact $\iff L$ is finite.*

Proof. $(\Rightarrow)$ If $|L|$ is compact, then it has intersection with finitely many open simplices. On the other hand, $|L| = \sqcup_{s \in L} \langle s \rangle$ so L is finite.

$(\Leftarrow)$ If L is finite, then $|L|$ is the finite union of its closed simplex, each of which is compact. The finite union of compact sets is also compact. Thus $|L|$ is compact. $\square$

Corollary 4.6. *$|K|$ is compact $\iff K$ is finite.*

The last thing in this section is about functoriality of $|\cdot|$ and $|\cdot|_d$.

Proposition 4.3. *$|\cdot|$ and $|\cdot|_d$ give two covariant functors from $\text{SC} \rightarrow \text{Top}$.*

Proof. For any given simplicial complex K , $|K|$ and $|K|_d$ are two topological spaces. For a simplicial map $K \rightarrow L$, we need to define what $|\varphi| : |K| \rightarrow |L|$ and $|\varphi|_d : |K|_d \rightarrow |L|_d$ are. We only need to define what $|\varphi|(\alpha)$ (equivalently $|\varphi|_d(\alpha)$) is for any $\alpha \in |K|$. Let $|\varphi|(\alpha) = |\varphi|_d(\alpha) := \sum_{v' \in V_L} (\sum_{\varphi(v)=v'} \alpha(v)) \chi_{\{v'\}}$. Or equivalently, $|\varphi|(\alpha)(v') = |\varphi|_d(\alpha)(v') = \sum_{\varphi(v)=v'} \alpha(v)$, $v' \in V_L$. This definition may not be that straightforward for the reader, but actually $|\varphi| = |\varphi|_d$ is "piecewise linear" (I have not define this terminology so far. I will define it in next section). To be specific, if $\alpha = \sum_{v \in V_K} \alpha(v) \chi_{\{v\}}$, then $|\varphi|(\alpha) = \sum_{v' \in V_L} (\sum_{\varphi(v)=v'} \alpha(v)) \chi_{\{v'\}} = \sum_{v \in V_K} \alpha(v) \chi_{\{\varphi(v)\}}$. The reader could easily verify that the image of $|\varphi|$ (or equivalently $|\varphi|_d$) in $|L|$, i.e.,

$$(1). \sum_{\varphi(v)=v'} \alpha(v) \in [0, 1], \forall v' \in L;$$

$$(2). \sum_{v' \in V_L} (\sum_{\varphi(v)=v'} \alpha(v)) = 1;$$

$$(3). \text{supp}(|\varphi|(\alpha)) \in L$$

(1) and (2) are straightforward. For (3) please prove $\text{supp}(|\varphi|(\alpha)) = \varphi(\text{supp}(\alpha)) \in L$.

Next we need to prove $|\varphi|$ and $|\varphi|_d$ are continuous. For $|\varphi|_d$,

$$|\varphi|_d(\alpha) = \sum_{v' \in V_L} \left(\sum_{\varphi(v)=v'} \alpha(v) \right) \chi_{\{v'\}}$$

$$|\varphi|_d(\beta) = \sum_{v' \in V_L} \left(\sum_{\varphi(v)=v'} \beta(v) \right) \chi_{\{v'\}}$$

Therefore

$$\begin{aligned}
d(|\varphi|_d(\alpha), |\varphi|_d(\beta)) &= \sum_{v' \in V_L} \left| \sum_{\varphi(v)=v'} \alpha(v) - \sum_{\varphi(v)=v'} \beta(v) \right| \\
&= \sum_{v' \in V_L} \left| \sum_{\varphi(v)=v'} (\alpha(v) - \beta(v)) \right| \\
&\leq \sum_{v' \in V_L} \sum_{\varphi(v)=v'} |\alpha(v) - \beta(v)| \\
&= \sum_{v \in V_K} |(\alpha(v) - \beta(v))| = d(\alpha, \beta)
\end{aligned}$$

Therefore $|\varphi|_d$ is not only continuous, it is Lipschitz continuous.

Finally let us deal with the continuity of $|\varphi|$. We only need to verify that the restriction map on $|s|$, i.e., $|\varphi|_{|s|} : |s| \rightarrow |L|$, is continuous. Note that $|\varphi|(|s|) = |\varphi(s)|$ (try to prove it on your own. One direction is simple and the other direction needs some construction but is not hard). Therefore for each $s \in K$, $|\varphi| : |s| \rightarrow |L|$ can be viewed as the composition of $|s| \rightarrow |\varphi|(|s|) = |\varphi(s)|$ and the inclusion map $|\varphi(s)| \hookrightarrow |L|$. The first map is continuous because the topology in $|s|$ and $|\varphi|$ is metric topology and $|\varphi|$ is Lipschitz continuous. The second map is continuous due to the property of weak topology.

By the discussion above, we have the following commutative diagram:

$$\begin{array}{ccc}
|K| & \xrightarrow{i(K)} & |K|_d \\
\downarrow |\varphi| & & \downarrow |\varphi|_d \\
|L| & \xrightarrow{i(L)} & |L|_d
\end{array}$$

where $i : |\cdot| \rightarrow |\cdot|_d$ is the identical natural transformation. □

The last definition will be used in the next section, but it is better to be introduced here.

Definition 4.4. A **triangulation** (K, f) of a topological space X consists of a simplicial complex K and a homeomorphism $f : |K| \rightarrow X$. If X has a triangulation, then X is called a **polyhedron**. There is also a pair version: a **triangulation** $((K, L), f)$ of a pair (X, A) consists of a simplicial pair (K, L) and a homeomorphism $f : (|K|, |L|) \rightarrow (X, A)$. If (X, A) has a triangulation, then (X, A) is called a **polyhedron pair**.

Remark 4.3. It is a difficult question whether a topological space is a polyhedron. In fact, all smooth manifolds are triangulable and algebraic varieties are triangulable. Extensive work is required to prove this result. Please check relevant papers if you are interested.

Example 4.3. $(D^{n+1}, S^n) \cong (|\bar{s}|, |\dot{s}|)$ where s is an $n+1$ -simplex, $n \geq 0$.

5 Subdivision

The goal of this section is to prove this theorem:

Theorem 5.1. *Suppose (X, A) is a polyhedral pair. Then A is a strong deformation retract of some neighborhood of it in X .*

To achieve this goal, a very important concept will be introduced. Given a simplicial complex K , its realization can be regarded as the union of a bunch of "higher dimensional triangles". Each triangle can be divided into the union of smaller "triangles". This operation is called subdivision and we will make it mathematically rigorous. After that, a specific subdivision, barycentric subdivision, will be introduced. Then we will construct this strong deformation retract by barycentric subdivision. This section corresponds to Chapter 3.3 of Spanier. Please note that the notation is different from Spanier. I also think that Spanier's narrative is not that clear so I will try to make it clearer. To begin with, let us prove an easy lemma:

Lemma 5.1. *Let K be a simplicial map and $s \in K$. Then $|s|$ has a linear structure on it. That is, given finitely many points in $|s|$, then the convex combination of these points also lies in $|s|$.*

Proof. Suppose $s = \{v_0, \dots, v_q\}$ and these points are $\{w_0, \dots, w_k\}$. The linear combination of these points is defined as $\sum_{j=0}^k r_j w_j$ where $\sum_{j=0}^k r_j = 1$ and $0 \leq r_j \leq 1$ for $0 \leq j \leq k$. Each v_i can be expressed by the convex combination of characteristic functions of the vertices of s , i.e.,

$$w_j = \sum_{i=0}^q t_{ji} \chi_{\{v_i\}}$$

where $\sum_{i=0}^q t_{ji} = 1, 0 \leq j \leq k$. and $0 \leq t_{ji} \leq 1$. Therefore,

$$\sum_{j=0}^k r_j w_j = \sum_{j=0}^k r_j \sum_{i=0}^q t_{ji} \chi_{\{v_i\}} = \sum_{i=0}^q \left(\sum_{j=0}^k t_{ji} r_j \right) \chi_{\{v_i\}}$$

which is apparently a convex combination of the characteristic functions of vertices of s , so we get the result. $\square$

Typically $\chi_{\{v\}}$ and v can be viewed as equivalent, so we can replace $\chi_{\{v\}}$ by v . Based on this convention, let K and K' be two simplicial complexes. A map $V_{K'} \xrightarrow{\psi} |K|$ induces a map $|K'| \xrightarrow{\Psi_\psi} \mathbb{R}^{\oplus V_K}$:

$$\Psi_\psi(\alpha) = \Psi_\psi\left(\sum_{v' \in V_{K'}} \alpha(v') \chi_{\{v'\}}\right) := \sum_{v' \in V_{K'}} \alpha(v') \psi(v')$$

$\mathbb{R}^{\oplus V_K}$ is the real vector space generated by V_K where the equivalence convention is used. Use the convention again to replace $\chi_{\{v'\}}$ by v' :

$$\Psi_\psi\left(\sum_{v' \in V_{K'}} \alpha(v') v'\right) = \sum_{v' \in V_{K'}} \alpha(v') \psi(v')$$

We call Ψ_ψ the **piecewise linear map** induced by ψ . One can observe that $|K| \subset \mathbb{R}^{\oplus V_k}$. However, we want the codomain of Ψ_ψ to be $|K|$, not $\mathbb{R}^{\oplus V_k}$, so a natural question would be: when Ψ_ψ is a map from $|K'|$ to $|K|$? The following proposition gives a necessary and sufficient condition:

Proposition 5.1. $\Psi_\psi(|K'|) \subset |K| \Leftrightarrow \forall s' \in K', \exists s \in K, \text{ s.t. } \psi(s') \subset |s|$.

Proof. $\Leftarrow$: For any $\alpha \in |K'|$, $\Psi_\psi(\alpha) = \sum_{v' \in V_{K'}} \alpha(v')\psi(v')$. Consider $s' = \text{supp}(\alpha) \in K'$, by the assumption $\exists s \in K \text{ s.t. } \psi(s') \subset |s|$. Therefore $\Psi_\psi(\alpha) = \sum_{v' \in V_{K'}} \alpha(v')\psi(v') = \sum_{v' \in s'} \alpha(v')\psi(v')$. This is the convex combination of $\{\psi(v') | v' \in s'\}$ and each $\psi(v')$ lies in $|s|$ so the convex combination will also lie in $|s|$ by lemma 5.1.

$\Rightarrow$: Suppose $s' \in K', s' = \{v_0, \dots, v_q\}$. Then $\Psi_\psi(|K'|) \subset |K| \Rightarrow \Psi_\psi(|s'|) \subset |K| \Leftrightarrow \{\sum_{i=0}^q t_i \psi(v'_i) | \sum_{i=0}^q t_i = 1, 0 \leq t_i \leq 1, 0 \leq i \leq q\} \subset |K|$. Choose $t_i = \frac{1}{1+q}$ for all i , then $\frac{1}{1+q} \sum_{i=0}^q \psi(v'_i) \in |K|$. Let $s_i := \text{supp}(\psi(v'_i))$ and $s = \cup_{i=0}^q s_i$. One can simply prove $s = \text{supp}(\frac{1}{1+q} \sum_{i=0}^q \psi(v'_i))$ and thus implies $s \in K$. Therefore, for all $0 \leq i \leq q$, $s_i = \text{supp}(\psi(v'_i)) \subset \cup_{i=0}^q \text{supp}(\psi(v'_i)) = s \in K$, which is exactly the definition of $\psi(v'_i) \in |s|$. $\square$

We can actually do better when we prove the necessity of the last proposition. We choose $t_i = \frac{1}{1+q}$ for each $0 \leq i \leq q$ to find the objective carrying simplex $s \in K$. We can actually generalize the subsequent proof to $\text{supp}(\sum_{i=0}^q t_i \psi(v'_i)) = s$ if $0 < t_i < 1$ for all $0 \leq i \leq q$. This immediately leads to the following corollary:

Corollary 5.1. *If $\Psi_\psi(|K'|) \subset |K|$, we can choose $s \in K$ such that $\Psi_\psi(\langle s' \rangle) \subset \langle s \rangle$. And since $|K|$ is the disjoint union of its open simplices, the choice of s is unique.*

By the discussion above, we can immediately build a bijection between the following two sets:

$$\{V_{K'} \xrightarrow{\psi} |K| \mid \forall s' \in K', \exists s \in K, \text{ s.t. } \psi(s') \subset |s|\} \Leftrightarrow \{|K'| \xrightarrow{\Psi} |K| \mid \Psi \text{ is piecewise linear map}\}$$

The correspondence from the left to the right is the induced Ψ_ψ . The correspondence from the right to the left is the restriction on the vertex set of K . One can check that they are bijective. Now we can give the first important definition of subdivision:

Definition 5.1. *A piecewise linear map $|K'| \xrightarrow{\Psi} |K|$ (or $V_{K'} \xrightarrow{\psi} |K|$) is a subdivision (of $|K|$) if Ψ is a homeomorphism.*

One can easily verify that the composition of two subdivisions is still a subdivision. The following theorem gives a sufficient and necessary condition when a piecewise linear map is a subdivision.

Theorem 5.2. *A piecewise linear map $\Psi : |K'| \rightarrow |K|$ is a subdivision $\iff \forall s \in K$, we have*

$$\langle s \rangle = \bigsqcup_{\substack{s' \in K' \\ \Psi(\langle s' \rangle) \subset \langle s \rangle \\ \text{finite union}}} \Psi(\langle s' \rangle)$$

*and the union is **finite** and **non-overlapping**.*

Proof. $(\Rightarrow)$: On the one hand, we have known that $\forall s' \in K', \exists! s \in K$ s.t. $\Psi(\langle s' \rangle) \subset \langle s \rangle$. On the other hand, $\forall s \in K, \Psi^{-1}(|s|)$ is compact because Ψ is a homeomorphism and hence there are at most finitely many $s' \in K'$ s.t. $\Psi(\langle s' \rangle) \subset |s|$.

$(\Leftarrow)$: Suppose a piecewise linear map Ψ satisfies

$$\langle s \rangle = \bigsqcup_{\substack{s' \in K' \\ \Psi(\langle s' \rangle) \subset \langle s \rangle \\ \text{finite union}}} \Psi(\langle s' \rangle)$$

where the union is a finite disjoint union. From this we can see that Ψ is a bijection. Apparently Ψ is a surjection. For injection, let us choose a point $\alpha \in |K|$. Then α lies in a unique open simplex $\langle s \rangle$. Given that the union is disjoint, there exists a unique $s' \in K'$ such that $\alpha \in \Psi(\langle s' \rangle)$. We only need to check that the preimage of α is a single point in $\langle s' \rangle$. If not, let us suppose we have two different points in $\langle s' \rangle$ whose image is α . Then let us connect these two points with a line segment and extend the line segment to the boundary of $|s'|$. All the points on that line is mapped to α by Ψ due to linearity, and so is the intersection point at the boundary. However, that point no longer belongs to $\langle s' \rangle$, so it is a contradiction, since the union is no longer disjoint. Hence Ψ is a bijection.

Next, we need to prove that Ψ is continuous and its inverse is also continuous. Similar to the proof of Theorem 4.3, we know that the piecewise linear map is continuous. Then we only need to verify its inverse is also continuous. This is where we use the condition that the disjoint union is finite. $\forall s \in K$, let us define $K'(s) := \{s' \in K' \mid \exists s_1 \subset s \text{ s.t. } \Psi(\langle s' \rangle) \subset \langle s_1 \rangle\} = \{s' \in K' \mid \Psi(\langle s' \rangle) \subset |s|\}$ (check the equivalence between the sets by definition on your own. Should be easy). Now I claim that $K'(s)$ is a finite subcomplex of K' . To verify this, we have to verify that $\forall s'' \subset s' \in K(s)$ implies $s'' \in K(s)$. This is because $\emptyset \neq s'' \subset s' \in K'(s) \Rightarrow \Psi(\langle s'' \rangle) \subset \Psi(|s''|) \subset \Psi(|s'|) = \Psi(\langle s' \rangle) \subset \Psi(\langle s' \rangle) \subset \overline{|s|} = |s|$. Hence $s'' \in K'(s)$ and $K'(s)$ is a subcomplex. $K'(s)$ is finite because for each $s \in K$ the corresponding open simple $\langle s \rangle$ is the finite disjoint union of the image of some open simplices $s' \in K'$ and $|s|$ can be written as the disjoint union of finitely many open simplices in K .

Since that $\Psi : |K'| \rightarrow |K|$ is a bijection, its restriction on $|K'(s)| \rightarrow |s|$ is also a bijection. Next, let us prove the composition of maps $|s| \xrightarrow{\Psi^{-1}} |K'(s)| \hookrightarrow |K'|$ is continuous, which directly leads to the result that Ψ^{-1} is continuous. This is because $|K'(s)|$ is compact by lemma 4.5. We also know $|s|$ is Hausdorff. A continuous bijection from a compact set to a Hausdorff set is a homeomorphism, so $|s| \xrightarrow{\Psi^{-1}} |K'(s)|$ is a homeomorphism, and the inclusion from $|K'(s)|$ to $|K'|$ is continuous due to the property of subset topology. Therefore $\Psi : |K'| \rightarrow |K|$ is an homeomorphism and a subdivision. $\square$

Corollary 5.2. *Let K be a simplicial complex. $L \subset K$ is a subcomplex, and $|K'| \xrightarrow{\Psi} |K|$ is a subdivision. Then $\exists!$ subcomplex $L' \subset K'$ s.t. $\Psi|_{|L'|} : |L'| \rightarrow |L|$ is a subdivision where $|L|$ and $|L'|$ have weak topology or subspace topology.*

Proof. Let $L' := \{s' \in K' \mid \Psi(\langle s' \rangle) \subset |L|\}$ and it is a subcomplex of K' (same proof as theorem 5.2). Therefore $\Psi|_{|L'|} : |L'| \rightarrow |L|$ is a piecewise affine map. $\forall s \in L \subset K$, we have

$$\langle s \rangle = \bigsqcup_{\substack{s' \in K' \\ \Psi(\langle s' \rangle) \subset \langle s \rangle \\ \text{finite union}}} \Psi(\langle s' \rangle) = \bigsqcup_{\substack{s' \in L' \\ \Psi(\langle s' \rangle) \subset \langle s \rangle \\ \text{finite union}}} \Psi(\langle s' \rangle)$$

This shows that $\Psi|_{|L'|}$ is a bijection thus homeomorphism with subspace topology or weak topology. $\square$

Definition 5.2. We call L' obtained by corollary 5.2 the induced subdivision (of L) from $|K'| \xrightarrow{\Psi} |K|$ and denote it as $K'|_L$.

Corollary 5.3. X is a topological space and $A \subset X$. $((K, L), f)$ is a triangulation of (X, A) . $|K'| \xrightarrow{\Psi} |K|$ is a subdivision. Then $((K', K'|_L), f \circ \Psi)$ is a triangulation of (X, A) too.

Before we proceed to introduce barycentric subdivision, the following lemma is necessary.

Lemma 5.2. K and K' are two simplicial complexes and $s \in K$. $V_{K'} \xrightarrow{\psi} |\overset{\circ}{s}|$ is a subdivision. Then $\forall p \in \langle s \rangle$, we can construct $\tilde{\psi} : V_K \sqcup \{*\} \rightarrow |\bar{s}| = |s|$

$$\tilde{\psi}(v) = \begin{cases} \psi(v) & \text{if } v \in V_{K'} \\ p & \text{if } v = *. \end{cases}$$

$\tilde{\psi}$ induces a subdivision $\Psi_{\tilde{\psi}} : |K' * \{*\}| \rightarrow |\bar{s}|$. Review example 3.4 for the definition of join if you don't remember it.

Proof. First of all, we need to prove that $\tilde{\psi}$ actually induces a piecewise linear map from the join $|K' * \{*\}|$ to $|\bar{s}|$. A simplex t in the join $|K' * \{*\}|$ has three possibilities: (a). $t \in K'$; (b). $t = \{*\}$; (c). $t = t' \cup \{*\}$ where $t' \in K'$. Since $\tilde{\psi}$ maps all points to $|s|$, by proposition 5.1 $\Psi_{\tilde{\psi}}$ is a piecewise linear map.

Next, we need to prove $\Psi_{\tilde{\psi}}$ is a homeomorphism by theorem 5.2. $\square$

Now let us introduce the concept of barycentric subdivision.

Definition 5.3. K is a simplicial complex. Its barycentric subdivision $\text{Sd } K$ is defined as $V_{\text{Sd } K} = \{b(s) \mid s = \{v_0, v_1, \dots, v_k\} \in K, b(s) = \frac{1}{k+1} \sum_{i=0}^k v_i \in |K|\}$ and $\{b(s_0), \dots, b(s_q)\} \in \text{Sd } K$ if $s_0 \subset s_1 \subset \dots \subset s_q$

Theorem 5.3. Let $\psi : V_{\text{Sd } K} \rightarrow |K|, \psi(b(s)) = b(s)$. Then ψ is a subdivision.

It is very clear that $\text{Sd } K$ is a simplicial complex according to definition. Next we will prove that there is a map ψ which is a subdivision of K .

Proof. First, let us prove that ψ is a piecewise linear map by using proposition 5.1. It is because for each $s = \{b(s_0), b(s_1), \dots, b(s_q)\} \in \text{Sd } K$, $\exists s_q$ such that $\psi(s) \in |s_q|$.

Next, let us prove ψ is a subdivision by theorem 5.2. For any $s \in K$, we collect all the simplices in $\text{Sd } K$ whose image lies in $\langle s \rangle$, i.e., $\{s' \in \text{Sd } K \mid \Psi_{\psi}(\langle s' \rangle) \subset \langle s \rangle\}$. By definition it can be proven that $\{s' \in \text{Sd } K \mid \Psi_{\psi}(\langle s' \rangle) \subset \langle s \rangle\} = \{s' \in \text{Sd } K \mid \text{last vertex of } s' = b(s)\} = \{s' \in \text{Sd } \bar{s} \mid \Psi_{\psi}(\langle s' \rangle) \subset \langle s \rangle\}$. In this case, if we can prove that $\text{Sd } \bar{s}$ is a subdivision of $\bar{s}$ for any $s \in K$, then that means the open simplices in the set $\{s' \in \text{Sd } \bar{s} \mid \Psi_{\psi}(\langle s' \rangle) \subset \langle s \rangle\}$ are disjoint and their union is $\langle s \rangle$. Since $s' \in \text{Sd } K$ as well, so the theorem is proven.

To prove that $\text{Sd } \bar{s}$ is a subdivision of $\bar{s}$, let us prove by induction. Apparently, if $\dim s = 0$, then $\text{Sd } \bar{s}$ is a subdivision for any $s \in K$. If $\dim s = q > 0$, then let us assume the statement holds for all $s_1 \in K$ where $\dim s < q$. Therefore $\text{Sd } \overset{\circ}{s}$ is a subdivision of $\overset{\circ}{s}$. By definition, $\text{Sd } \bar{s} = \text{Sd } \overset{\circ}{s} * b(s)$. By lemma 5.2, this is a subdivision of $\bar{s}$. $\square$

We are fairly close to the objective of this section, that is, to prove theorem 5.1. Before this, we introduce some concepts which will facilitate the construction of the "open neighborhood" of $|L|$. Let $N := \{s \in K \mid s \cap V_L = \emptyset\}$. This is the largest subcomplex of K which is disjoint from L . Let us take a look at some examples (Fig. 1) of the relationship between $|L|$ and $|N|$ in $|K|$ and see what we can get from them.

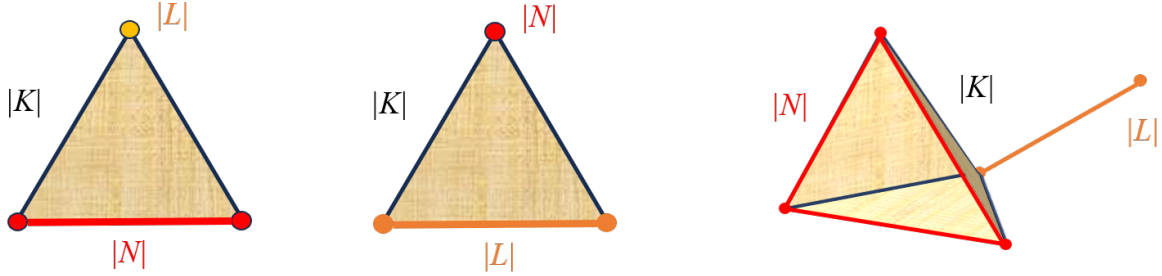


Figure 1: some examples

These three examples have something in common. Since $|N|$ is a closed set in $|K|$, $|K| - |N|$ is open in $|K|$. Therefore, they are open neighborhoods of $|L|$, and $|L|$ are strong deformation retract of $|K| - |N|$ in all three examples. Therefore, we want to ask the following question: is $|L|$ always the strong deformation retract of $|K| - |N|$?

Unfortunately, the answer is negative. Fig. 2(a) shows a counter example. Apparently $|K| - |N|$ cannot be retracted to $|L|$. However if we subdivide K , then $|K| - |N|$ can be retracted to $|L|$, see 2(b). So another question is: if $|L|$ is not the strong deformation retract of $|K| - |N|$, can we do barycentric subdivision so $|L|$ becomes the deformation retract of $|\text{Sd } K| - |N|$? The answer is positive and the proof is given in Lemma 5.3. Before the proof, we summarize the property and give a name to those L if its realization is the strong deformation retract of $|K| - |N|$.

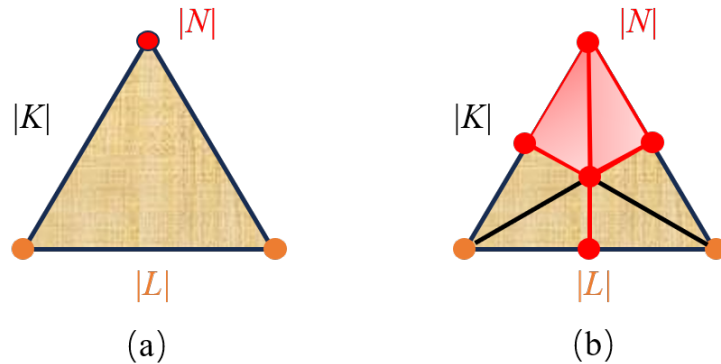


Figure 2: a counter example

Definition 5.4. L is a full subcomplex of K if $s = \{v_0, \dots, v_q\} \in K$ and $v_0, \dots, v_q \in V_L$, then $s \in L$.

We can see that L in Fig. 1 are full subcomplexes, but L in Fig.2(a) is not a full complex. However, if you add the barycenters to K , then L becomes a full subcomplex.

Lemma 5.3. (a). $\text{Sd } L = \text{Sd } K|_L$; (b). $\text{Sd } L$ is a full subcomplex of $\text{Sd } K$.

Proof. (a). Let us prove $\text{Sd } L \subset \text{Sd } K|_L := \{s' \in \text{Sd } K \mid \Psi_\psi(\langle s' \rangle) \subset |L|\}$ first. Let $s = \{b(s_0), \dots, b(s_q)\} \in \text{Sd } L$, $s_0 \subsetneq s_1 \subsetneq \dots \subsetneq s_q$, $s_0, \dots, s_q \in L$. It is obvious that $s \in \text{Sd } K$ given that $s_0, \dots, s_q \in K$. We need to prove that $\Psi_\psi(\langle s \rangle) \subset |L|$. This is also obvious because $\psi(s) = \{b(s_0), \dots, b(s_q)\} \subset |s_q| \subset |L|$, then by Lemma 5.1, the convex combination of $\{b(s_0), \dots, b(s_q)\}$ is also a subset of $|s_q| \subset |L|$, so $\Psi_\psi(\langle s \rangle) \subset |s_q| \subset |L|$.

Next, let us prove $\text{Sd } L \supset \text{Sd } K|_L := \{s' \in \text{Sd } K \mid \Psi_\psi(\langle s' \rangle) \subset |L|\}$. Let $s = \{b(s_0), \dots, b(s_q)\} \in \text{Sd } K$ such that $\Psi_\psi(\langle s \rangle) \subset |L|$. We have $\Psi_\psi(|s|) = \Psi_\psi(\langle s \rangle) \subset \overline{\Psi_\psi(\langle s \rangle)} \subset \overline{|L|} = |L| \Rightarrow \Psi_\psi(\chi_{b(s_q)}) = \psi(b(s_q)) = b(s_q) \in |L|$. Similar to the proof of necessity of Proposition 5.1, $b(s_q) \in |L|$ implies $s_q \in L$. Specifically, let $s_q = \{v_0, \dots, v_n\} \in K$, so $b(s_q) = \frac{1}{n+1} \sum_{i=0}^n \chi_{\{v_i\}}$. We also have $\text{supp}(\chi_{\{v_i\}}) = \{v_i\}$. It can be proved that $\text{supp}(b(s_q)) = \cup_{i=0}^n \{v_i\} = \{v_0, \dots, v_n\} = s_q$. Since $b(s_q) \in |L|$, by Proposition 4.2, $\cup_{i=0}^n \{v_i\} = s_q \in L$ and by definition of subcomplex, $s_0 \subsetneq \dots \subsetneq s_q$ are all in L .

(b). Let $s = \{b(s_0), \dots, b(s_q)\} \in \text{Sd } K$ ($s_0 \subsetneq \dots \subsetneq s_q$) and $b(s_i) \in V_{\text{Sd } L}, \forall 0 \leq i \leq q$. We want to prove that $s \in \text{Sd } L$. Denote $s_i = \{v_{i,0}, \dots, v_{i,n_i}\} \in K$ and $b(s_i) = \frac{1}{n_i+1} \sum_{j=0}^{n_i} v_{i,j} \in |K|$. If we can conclude $s_i \in L, \forall 0 \leq i \leq q$, then it directly implies $s \in \text{Sd } L$. To conclude this, we know that for each s_i , $b(s_i) \in |L|$ because $b(s_i) \in V_{\text{Sd } L}$ is the barycenter of some simplex $s' \in L$. So $\text{supp}(b(s_i)) = s' = \cup_{j=0}^{n_i} \text{supp}(\chi_{\{v_{i,j}\}}) = \{v_{i,0}, \dots, v_{i,n_i}\}$. The second and the third equalities use the same technique in (a). $\square$

Lemma 5.4. L is a full subcomplex of K and $N := \{s \in K \mid s \cap V_L = \emptyset\}$. Then N is also full and for any $s \in K$, $s = s' \sqcup s''$ such that $s' \in L$ and $s'' \in N$. Of course, s' and s'' can be empty sets.

Proof. By definition of N , N is obviously full. For any $s \in K$ there are three cases: (a). if $s \subset V_L$, then $s \in L$. (b). if $s \subset V_N$ then $s \in N$. (c). if $s = s' \sqcup s''$ s.t. $s' \in V_L, s'' \in V_N$, then by the definition of full subcomplex $s' \in L$ and $s'' \in N$. $\square$

Now let us begin the proof of the core theorem - Theorem 5.1. There are some technical parts in the proof. The reader is recommended to read the main part of the proof, and just take the technical part as a reference, if time is limited, since the technical part is some verification of general topology and can be too tedious and lengthy.

Proof. We may assume that $(X, A) = (|K|, |L|)$ where L is a **full** subcomplex of a simplicial complex K . The reason we can assume L is full is if not we can do barycentric subdivision of K , so $\text{Sd } L$ becomes a full subcomplex and $(\text{Sd } K, \text{Sd } L)$ is homeomorphic to (X, A) . We will construct a **continuous** map

$$(|K| - |N| \times [0, 1]) \xrightarrow{F} |K| - |N|$$

which provides a homotopy between identity and the retract $|K| - |N| \xrightarrow{r} |L| \hookrightarrow |K| - |N|$. $\forall \alpha \in |K| - |N|$, let $s = \{v \in K \mid \alpha(v) \neq 0\}$, or equivalently, $\alpha \in \langle s \rangle$. By Lemma 5.4,

$s = s' \sqcup s''$, $s = \{v_0, \dots, v_p\} \in L$ and $s'' = \{v_{p+1}, \dots, v_q\} \in N$ (if not empty). Therefore, $\alpha = \sum_{i=0}^q \alpha(v_i)v_i, \alpha(v_i) > 0$. Let $\alpha' := \frac{\sum_{i=0}^p \alpha_i v_i}{\sum_{i=0}^p \alpha_i} \in |L|$ and $\alpha'' := \frac{\sum_{i=p+1}^q \alpha_i v_i}{\sum_{i=p+1}^q \alpha_i} \in |N|$. Figure 3 shows the position of these points. Let $F(\alpha, t) := (1-t)\alpha + t\alpha', t \in [0, 1]$. Therefore, $F : (|s| - |N|) \times [0, 1]$ is a **continuous** function. Now comes the technical part. There are two claims. Claim 1: F can be continuously extended to $(|s| - |N|) \times [0, 1]$. Claim 2: The product topology on $(|K| - |N|) \times [0, 1]$ coincides with the weak topology induced by $\{(|s| - |N|) \times [0, 1] \mid s \in K, |s| - |N| \neq \emptyset\}$. Given these two claims and the Proposition 2.1, $F : (|K| - |N|) \times [0, 1]$ is continuous.

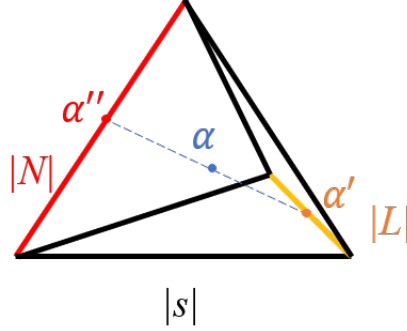


Figure 3: position of α , α' and α''

The reader who is not interested in the proof of both claims can take them as granted and skip the rest of the proof now.

Proof of claim 1: This claim is actually very straightforward to prove. Let $\alpha \in |s| - |N|$ so $\alpha = \sum_{i=0}^n \alpha(v_i)v_i$. Given $\alpha \notin |N|$, $\alpha(v_i), p+1 \leq i \leq q$ cannot be all 0. Therefore $\alpha' := \frac{\sum_{i=0}^p \alpha_i v_i}{\sum_{i=0}^p \alpha_i}$ is always well-defined and α' is continuously dependent on α . The linear combination of α and α' is also continuous so F is continuously extended to $(|s| - |N|) \times [0, 1]$.

Proof of claim 2: the proof of this claim is a lengthy one and not included in Spanier chapter 3.3 (or maybe it appears somewhere else and I didn't find it). Now let us prove:

the product topology on $|K| \times [0, 1]$ ($|K|$ equips with weak topology) coincides with the weak topology induced by $\{|s| \times [0, 1] \mid s \in K\}$ ($|s| \times [0, 1]$ equips with product topology).

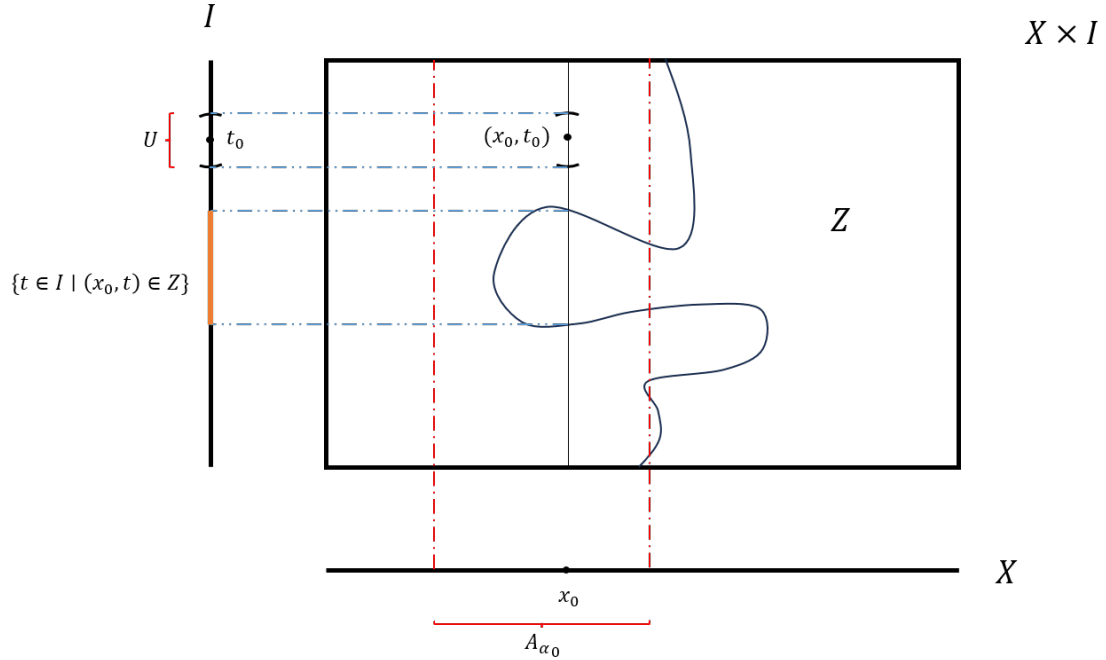
In fact, it is unlikely that you can find the proof of this in any textbook. Once this got proved, then the subspace topology on $(|K| - |N|) \times [0, 1]$ induced from the product topology $|K| \times [0, 1]$ coincides with the weak topology induced by $\{(|s| - |N|) \times [0, 1] \mid s \in K, |s| - |N| \neq \emptyset\}$. To do this, simply verify the closed sets are identical in both topologies which should be straightforward. After that, since the product topology of two subspaces is identical with the subspace topology of the product, claim 2 got proved. Now let us prove the following lemma.

Lemma 5.5. *Let X be a set and $\{A_\alpha \mid \alpha \in \mathcal{A}\}$ is a covering of X . Every of those members has a specified topology and is compact and Hausdorff. X is locally compact Hausdorff. Then*

the product topology on $X \times I$ (X has weak topology induced by $\{A_\alpha \mid \alpha \in \mathcal{A}\}$) coincides with the weak topology induced by $\{A_\alpha \times I \mid \alpha \in \mathcal{A}\}$.

Proof. It is clear that the weak topology on $X \times I$ is finer than the product topology. This is because weak topology is the finest topology to make $A_\alpha \times I \hookrightarrow X \times I$ continuous for all $\alpha \in \mathcal{A}$. Therefore, it suffices to show that if $Z \subset X \times I$ such that $Z \cap (A_\alpha \times I)$ is closed in $A_\alpha \times I$ (with product topology) for all $\alpha \in \mathcal{A}$, then Z is closed in $X \times I$ (with product topology). Equivalently, we need to prove $X \times I - Z$ is open in $X \times I$ with product topology.

For any $(x_0, t_0) \in X \times I - Z$, choose $\alpha_0 \in \mathcal{A}$ such that $x_0 \in A_{\alpha_0}$. Consider $\{t \in I \mid (x_0, t) \in Z\}$ (labeled orange). See the figure below.



$\{t \in I \mid (x_0, t) \in Z\}$ is closed. This is because: 1. the injection $i_{x_0} : I \rightarrow A_{\alpha_0} \times I, i_{x_0}(t) = (x_0, t)$ is continuous; 2. $\{t \in I \mid (x_0, t) \in Z\} = i_{x_0}^{-1}(Z \cap (A_{\alpha_0} \times I))$, and $(A_{\alpha_0} \times I) \cap Z$ is closed. Given that $t_0 \notin \{t \in I \mid (x_0, t) \in Z\}$ and locally compact Hausdorff of I , we can choose an open neighborhood U of t_0 in I such that $\overline{U}$ is compact and does not intersect $\{t \in I \mid (x_0, t) \in Z\}$. Therefore, $(\{x_0\} \times \overline{U}) \cap Z = \emptyset$.

Let $W := \{x \in X \mid (\{x\} \times \overline{U}) \cap Z = \emptyset\}$. Apparently, $x_0 \in W$. I claim that W is open and if this is true, then we can find a open neighborhood of x_0 in X such that the product of this open neighborhood and U is an interior neighborhood of (x_0, t) in $X \times I - Z$, which is exactly our goal. To prove the claim, let us check the complement $X - W$ is closed in X , which by definition of weak topology, is equivalent to $(X - W) \cap A_\alpha$ is closed in A_α for all $\alpha \in \mathcal{A}$. Denote the projection map $\pi : X \times I \rightarrow X$, then $X - W = \pi((X \times \overline{U}) \cap Z)$. Therefore, $(X - W) \cap A_\alpha = \pi((X \times \overline{U}) \cap Z) \cap \pi(A_\alpha \times I) = \pi((X \times \overline{U}) \cap (A_\alpha \times I) \cap Z) = \pi((A_\alpha \times \overline{U}) \cap Z) = \pi((A_\alpha \times \overline{U}) \cap (A_\alpha \times I) \cap Z)$ and $A_\alpha \times \overline{U}$ is compact, $(A_\alpha \times I) \cap Z$ is closed in $A_\alpha \times I$, so $\pi((A_\alpha \times \overline{U}) \cap (A_\alpha \times I) \cap Z)$ is compact subset of Hausdorff space A_α , which is closed in A_α . $\square$

□

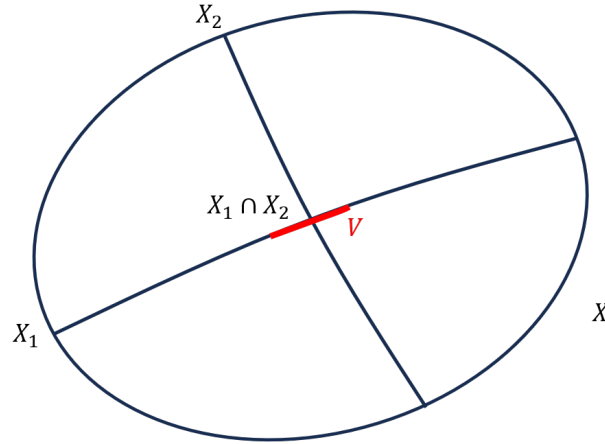
Corollary 5.4. *Let K be a simplicial complex. K_1 is a subcomplex of K and K_2 is a subcomplex of K_1 . Then $|K_2|$ is a strong deformation retract of an open neighborhood in $|K_1|$.*

Proof. Obviously, K_2 is a subcomplex of K . By Theorem 5.1, $|K_2|$ is a deformation retract of some open neighborhood in $|K|$. Therefore, its intersection with $|K_1|$ is open in $|K_1|$ and $|K_2|$ is a strong deformation retract of intersection. □

Lemma 5.6. *X is a topological space. X_1 and X_2 are closed subspaces of X . If $r : V \rightarrow X_1 \cap X_2$ is a strong deformation retract from the open neighborhood V of $X_1 \cap X_2$ in X_1 , then $V \cup X_2$ is an open neighborhood of X_2 in $X_1 \cup X_2$ and $\tilde{r} : V \cup X_2 \rightarrow X_2$,*

$$\tilde{r}(p) = \begin{cases} r(p) & p \in V \\ p & p \in X_2 \end{cases}$$

is a strong deformation retract. See the figure below for reference.



Proof. First, to prove $V \cup X_2$ is open in $X_1 \cup X_2$, we need to prove that $X_1 \cup X_2 - V \cup X_2 = X_1 - V$ is closed in $X_1 \cup X_2$. Of course, it is closed in X_1 , and X_1 is closed in X , so $X_1 - V$ is closed in X . By the definition of subspace topology, it is closed in $X_1 \cup X_2$.

Secondly, by gluing lemma, $\tilde{r}$ is continuous. Next, let us build the homotopy. Let $h : V \times [0, 1] \rightarrow V$ be homotopy map of the strong deformation retract, i.e., $h(p, 0) = p$, $h(p, 1) = r(p)$ for all $p \in V$ and $h(p, t) = p$ for all $p \in V \cap X_2$, $t \in [0, 1]$. Let $\tilde{h} : (V \cup X_2) \times [0, 1] \rightarrow V \cup X_2$:

$$\tilde{h}(p, t) = \begin{cases} h(p, t) & p \in V \\ p & p \in X_2 \end{cases}$$

By gluing lemma, $\tilde{h}$ is continuous and $\tilde{h}(p, 0) = p$; $\tilde{h}(p, 1) = \tilde{r}(p)$. □

Theorem 5.4. *X is a topological space, X_1 and X_2 are closed in X . $X_1 \cap X_2$ is a strong deformation retract of some open neighborhood of it in X_1 , then X_1 and X_2 form an excisive couple, i.e., $S_\bullet(X_1) + S_\bullet(X_2) \xrightarrow{i} S_\bullet(X_1 \cup X_2)$ induces an isomorphism between homologies $H_k(S_\bullet(X_1) + S_\bullet(X_2)) \xrightarrow{i_*} H_k(X_1 \cup X_2)$.*

Proof. Let us use the following theorem to prove Theorem 5.4: X_1 and X_2 form an excisive couple $\iff (X_1, X_1 \cap X_2) \hookrightarrow (X_1 \cup X_2, X_2)$ induces an isomorphism $H_k(X_1, X_1 \cap X_2) \rightarrow H_k(X_1 \cup X_2, X_2)$. In fact, we have the following long exact sequence:

$$\begin{array}{ccccccccccc} \cdots & \rightarrow & H_k(X_1 \cap X_2) & \rightarrow & H_k(X_1) & \rightarrow & H_k(X_1, X_1 \cap X_2) & \rightarrow & H_{k-1}(X_1 \cap X_2) & \rightarrow & H_{k-1}(X_1) & \rightarrow & \cdots \\ & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \\ \cdots & \longrightarrow & H_k(V) & \longrightarrow & H_k(X_1) & \longrightarrow & H_k(X_1, V) & \longrightarrow & H_{k-1}(V) & \longrightarrow & H_{k-1}(X_1) & \rightarrow & \cdots \end{array}$$

Similarly, we have

$$\begin{array}{ccccccccccc} \cdots & \longrightarrow & H_k(X_2) & \longrightarrow & H_k(X_1 \cup X_2) & \longrightarrow & H_k(X_1 \cup X_2, X_2) & \longrightarrow & H_{k-1}(X_2) & \longrightarrow & H_{k-1}(X_1 \cup X_2) & \rightarrow & \cdots \\ & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow & & \\ \cdots & \rightarrow & H_k(X_2 \cup V) & \rightarrow & H_k(X_1 \cup X_2) & \rightarrow & H_k(X_1 \cup X_2, V \cup X_2) & \rightarrow & H_{k-1}(X_2 \cup V) & \rightarrow & H_{k-1}(X_1 \cup X_2) & \rightarrow & \cdots \end{array}$$

All the vertical arrows are homomorphisms induced by inclusion. By 5-lemma and the previous lemma, $H_k(X_1, X_1 \cap X_2) \xrightarrow{i_*} H_k(X_1, V)$ and $H_k(X_1 \cup X_2, X_2) \xrightarrow{i_*} H_k(X_1 \cup X_2, V \cup X_2)$ are isomorphisms for all k (i is inclusion). On the other hand, $\{X_1 - X_2, V \cup X_2\}$ is an open cover of $X_1 \cup X_2$ and therefore, $\text{int}_{X_1 \cup X_2} X_1 \cup \text{int}_{X_1 \cup X_2} X_2 \cup V = X_1 \cup X_2$. This implies X_1 and $X_2 \cup V$ form an excisive couple, which is equivalent to $H_k(X_1, V) \rightarrow H_k(X_1 \cup X_2, V \cup X_2)$ is an isomorphism. Therefore, $H_k(X_1, X_1 \cap X_2) \xrightarrow{i_*} H_k(X_1 \cup X_2, X_2)$ is an isomorphism. $\square$

Remark 5.1. *In singular homology, if the interior of two sets are an open cover of the entire space, then they form an excisive couple. However, if we take S^2 as an example and choose the closed northern and southern hemisphere as X_1 and X_2 respectively, they do not satisfy this requirement. However, Theorem 5.4 tells us that they still form an excisive couple because we can choose an open neighborhood of the equator and the equator is the deformation retract of that open neighborhood.*

Corollary 5.5. *K is a simplicial complex and K_1 and K_2 are subcomplex of K . Then $|K_1|$ and $|K_2|$ form an excisive couple.*

Proof. $|K_1| \cap |K_2| = |K_1 \cap K_2|$ is a subcomplex of K_1 . By Corollary 5.4, it is a strong deformation retract of some open neighborhood in $|K_1|$. By the previous theorem, the corollary is concluded. $\square$

6 The Ordered Simplicial Homology

All the previous sections are introduction of simplicial homology and its realization. It has nothing to do with homology. In this section, a new homology theory is introduced.

Let k be a non-negative integer and $K[k] := \{s \mid \emptyset \neq s \subset \{1, 2, \dots, k\}\}$. So $|K[k]|$ is homeomorphic to the standard k simplex Δ^k . Let the **ordered simplicial chain complex** be

$$S_{\bullet}^{\text{ord}}(K) : \dots \longrightarrow S_k^{\text{ord}}(K) \xrightarrow{\partial_k^{\text{ord}}(K)} S_{k-1}^{\text{ord}}(K) \longrightarrow \dots \longrightarrow S_1^{\text{ord}}(K) \longrightarrow S_0^{\text{ord}}(K) \longrightarrow 0$$

where S_k^{ord} is a free abelian group generated by all simplicial maps from $K[k]$ to K . In this case, we can write each simplicial map as $[v_0, v_1, \dots, v_k]$ where $\{v_0, v_1, \dots, v_k\} \in K$. Note that $\{v_0, v_1, \dots, v_k\} \in K$ may have repetition so just leave one repetitive elements in the set. The reason why it is called ordered simplicial chain complex is that $[v_0, v_1, \dots, v_k]$ with different orders represent different simplicial maps. The boundary map $\partial_k^{\text{ord}}(K)$ is defined as

$$\partial_k^{\text{ord}}(K) = \sum_{i=0}^k (-1)^i [v_0, v_1, \dots, v_{i-1}, v_{i+1}, \dots, v_k]$$

and it satisfies $\partial_{k-1}^{\text{ord}}(K) \partial_k^{\text{ord}}(K) = 0$. The **ordered homology group** $H_k^{\text{ord}} := \text{Ker } \partial_k^{\text{ord}}(K) / \text{Im } \partial_{k+1}^{\text{ord}}(K)$. The same notion can be extended to pair (K, L) . In fact, $S_{\bullet}^{\text{ord}}(K)$ is much smaller than singular chain complex. If K has finite n -simplices for each non-negative integer n , then each $S_n^{\text{ord}}(K)$ is finitely generated. Moreover, if K is a finite complex, then $S_n^{\text{ord}}(K) = 0$ if n is sufficient large.

Like singular homology, ordered simplicial complex also has Mayer-Vietoris sequence. The proof for ordered simplicial complex version is much simpler than the singular version.

Theorem 6.1. *K is a simplicial complex and K_1 and K_2 are two subcomplexes. We know that $K_1 \cap K_2$ and $K_1 \cup K_2$ are also two subcomplexes of K . Then we have long exact sequence called Mayer-Vietoris sequence*

$$\dots \rightarrow H_n^{\text{ord}}(K_1 \cap K_2) \rightarrow H_n^{\text{ord}}(K_1) \oplus H_n^{\text{ord}}(K_2) \rightarrow H_n^{\text{ord}}(K_1 \cup K_2) \rightarrow H_{n-1}^{\text{ord}}(K_1 \cap K_2) \rightarrow \dots$$

Proof. Observe that $S_{\bullet}^{\text{ord}}(K_1) + S_{\bullet}^{\text{ord}}(K_2) = S_{\bullet}^{\text{ord}}(K_1 \cup K_2)$ by comparing the generators on both sides. Then we have short exact sequence of chain complex $0 \rightarrow S_{\bullet}^{\text{ord}}(K_1 \cap K_2) \rightarrow S_{\bullet}^{\text{ord}}(K_1) \oplus S_{\bullet}^{\text{ord}}(K_2) \rightarrow S_{\bullet}^{\text{ord}}(K_1 \cup K_2) \rightarrow 0$, which induces the long exact sequence of homology. $\square$

Example 6.1. *K is a simplicial complex and $s \in K$. Then the chain complex*

$$\dots \rightarrow S_k^{\text{ord}}(\bar{s}) \rightarrow S_{k-1}^{\text{ord}}(\bar{s}) \rightarrow \dots \rightarrow S_1^{\text{ord}}(\bar{s}) \rightarrow S_0^{\text{ord}}(\bar{s}) \rightarrow \mathbb{Z} \rightarrow 0$$

has zero homology.

Proof. Let us construct a homomorphism $h_k : S_k^{\text{ord}}(\bar{s}) \rightarrow S_{k+1}^{\text{ord}}(\bar{s})$, $k \geq -1$ ($S_{-1}^{\text{ord}}(\bar{s}) = \mathbb{Z}$). Recall that $S_k^{\text{ord}}(\bar{s})$ is freely generated by k -dimensional ordered simplices $\{[v_0, \dots, v_k]\}$ (or 1 when $k = -1$). If $k \geq n$, then $h_k = 0$. Otherwise, choose a vertex v in the vertex of $\bar{s}$ and let $h_k([v_0, \dots, v_k]) = [v, v_0, \dots, v_k]$ (For the case $k = -1$, $h_{-1}(1) = [v]$). It is straightforward to verify that $h_{k-1} \circ \partial_k + \partial_{k+1} \circ h_k = \text{id}$. Therefore, $\tilde{H}_k^{\text{ord}}(\bar{s}) = 0$ $\square$

Here is a very interesting question. What is the connection between ordered simplicial homology $H_k^{\text{ord}}(K)$ and singular homology $H_k(K)$? The answer is that they are isomorphic. To prove this, we will prove the case for all finite simplicial complexes and use a relatively advanced algebraic tool called direct limit to extend it to infinite cases.

To start with, we have a natural injective chain map (check by yourself that this is a chain map. You need to check the boundary map and the chain map are commutative) $\alpha_\bullet : S_\bullet^{\text{ord}}(K) \rightarrow S_\bullet(|K|)$, $\alpha_\bullet([v_0, v_1, \dots, v_k]) = \sigma$ where $\sigma : \Delta^k \rightarrow |K|$, $\sigma(t_0, t_1, \dots, t_k) = t_0 v_0 + t_1 v_1 + \dots + t_k v_k$. Therefore, $\alpha_\bullet$ induces a homomorphism between homology α_* . Now I claim:

Theorem 6.2. $\alpha_* : H_\bullet^{\text{ord}}(K) \rightarrow H_\bullet(|K|)$ is an isomorphism

Proof. Case 1: K is finite. Let us prove by induction on $\#K$.

(a). If $\#K = 1$, then obviously the claim is true.

(b). Assume the claim is true when $\#K \leq n-1$ and we want to extend the statement to the case of $\#K = n$. Let $s_0 \in K$ be of the top dimension and $L := \{s \in K \mid s \neq s_0\}$. (For example, if K represents a solid tetrahedron, then s_0 is the only 3rd dimensional simplex in K , and L are all other vertices and edges). By definition, L is a subcomplex and $\#L = n-1$. Note that $K = L \cup \overline{s_0}$. Then we have the following Mayer-Vietoris sequence:

$$\begin{array}{ccccccccc} \cdots & \rightarrow & H_k^{\text{ord}}(L \cap \overline{s_0}) & \rightarrow & H_k^{\text{ord}}(L) \oplus H_k^{\text{ord}}(\overline{s_0}) & \rightarrow & H_k^{\text{ord}}(K) & \rightarrow & H_{k-1}^{\text{ord}}(L \cap \overline{s_0}) \rightarrow \cdots \\ & & \downarrow \alpha_* & & \downarrow \alpha_* & & \downarrow \alpha_* & & \downarrow \alpha_* \\ \cdots & \rightarrow & H_k(|L \cap \overline{s_0}|) & \rightarrow & H_k(|L|) \oplus H_k(|\overline{s_0}|) & \rightarrow & H_k(|K|) & \rightarrow & H_{k-1}(|L \cap \overline{s_0}|) \rightarrow \cdots \end{array}$$

By induction hypothesis and five-lemma, $\alpha_* : H_k^{\text{ord}}(K) \rightarrow H_k(|K|)$ is an isomorphism.

Case 2: K is infinite. Then use direct limit. $\square$

7 The Oriented Simplicial Homology

Ordered simplicial homology is too redundant. In fact, we can create a more elegant homology theory from ordered simplicial homology which is simpler to be calculated - that is oriented simplicial homology. Normally in literatures simplicial homology is referred to as oriented simplicial homology, and ordered simplicial homology is just a middleware to prove the equivalence between simplicial homology and singular homology. In this section, we will introduce the oriented simplicial homology theory and prove its equivalence to singular homology.

Let K be a simplicial complex and k is a non-negative integer. We have defined $S_k^{\text{ord}}(K) := \mathbb{Z}^{\oplus \{[v_0, \dots, v_k] \mid \{v_0, \dots, v_k\} \in K\}}$. Let R_k be the subgroup of $S_k^{\text{ord}}(K)$ generated by $\{[v_0, \dots, v_k] - \text{sgn}(s)[v_{s(0)}, \dots, v_{s(k)}] \mid \{v_0, \dots, v_k\} \in K, s \in \text{Sym}(\{0, \dots, k\})\}$.

Lemma 7.1. $\partial_k(K)R_k \subseteq R_{k-1}$ for all $k \geq 0$.

Proof. We just need to prove this property on each generator of R_k .

$$\begin{aligned} & \partial_k(K)([v_0, \dots, v_k] - \text{sgn}(s)[v_{s(0)}, \dots, v_{s(k)}]) \\ &= \sum_{i=0}^k (-1)^i [v_0, \dots, \hat{v}_i, \dots, v_k] - \sum_{j=0}^k (-1)^j \text{sgn} \begin{pmatrix} 0 & \cdots & k \\ s(0) & \cdots & s(k) \end{pmatrix} [v_{s(0)}, \dots, \hat{v}_{s(j)}, \dots, v_{s(k)}] \end{aligned}$$

$\forall i \in \{0, \dots, k\}$, let $j = s^{-1}(i)$ and consider the permutation τ_i of $\{0, \dots, \hat{i}, \dots, k\}$:

$$\tau_i = \begin{pmatrix} 0 & \dots & \hat{i} & \dots & k \\ s(0) & \dots & s(\hat{j}) & \dots & s(k) \end{pmatrix}$$

Then

$$\begin{aligned} \text{sgn}(\tau_i) &= \text{sgn} \begin{pmatrix} i & 0 & \dots & \hat{i} & \dots & k \\ s(\hat{j}) & s(0) & \dots & s(\hat{j}) & \dots & s(k) \end{pmatrix} \\ &= \text{sgn} \begin{pmatrix} s(0) & \dots & s(j) & \dots & s(k) \\ s(\hat{j}) & s(0) & \dots & s(\hat{j}) & \dots & s(k) \end{pmatrix} \begin{pmatrix} 0 & \dots & k \\ s(0) & \dots & s(k) \end{pmatrix} \begin{pmatrix} i & 0 & \dots & \hat{i} & \dots & k \\ s(0) & \dots & s(j) & \dots & s(k) \end{pmatrix} \\ &= (-1)^i (-1)^j \text{sgn}(s) \end{aligned}$$

Therefore,

$$\begin{aligned} \partial_k(K)([v_0, \dots, v_k] - \text{sgn}(s)([v_{s(0)}, \dots, v_{s(k)}])) \\ &= \sum_{i=0}^k (-1)^i [v_0, \dots, \hat{v}_i, \dots, v_k] - \sum_{j=0}^k (-1)^j \text{sgn} \begin{pmatrix} 0 & \dots & k \\ s(0) & \dots & s(k) \end{pmatrix} [v_{s(0)}, \dots, \hat{v}_{s(j)}, \dots, v_{s(k)}] \\ &= \sum_{i=0}^k (-1)^i [v_0, \dots, \hat{v}_i, \dots, v_k] - \sum_{i=0}^k (-1)^{s^{-1}(i)} \text{sgn}(\tau_i) (-1)^i (-1)^{s^{-1}(i)} [v_{\tau_i(0)}, \dots, v_{\tau_i(i-1)}, v_{\tau_i(i+1)}, \dots, v_{\tau_i(k)}] \\ &= \sum_{i=0}^k (-1)^i ([v_0, \dots, \hat{v}_i, \dots, v_k] - \text{sgn}(\tau_i) [v_{\tau_i(0)}, \dots, v_{\tau_i(i-1)}, v_{\tau_i(i+1)}, \dots, v_{\tau_i(k)}]) \in R_{k-1} \end{aligned}$$

□

Therefore, we can construct a short exact sequence of chain complex $0 \rightarrow R_\bullet \rightarrow S_\bullet^{\text{ord}}(K) \rightarrow C_\bullet(K) \rightarrow 0$ where

$$C_\bullet(K) = S_\bullet^{\text{ord}}(K)/R_\bullet$$

Definition 7.1. $C_\bullet(K) : \dots C_k(K) \xrightarrow{\partial_k} C_{k-1}(K) \rightarrow \dots \rightarrow C_0(K) \rightarrow 0$ is called the oriented simplicial complex of K .

Let us denote the projection map $\mu_k : S_k^{\text{ord}}(K) \rightarrow C_k(K)$ and for each basis $[v_0, \dots, v_k] \in S_k^{\text{ord}}(K)$, denote $\mu_k([v_0, \dots, v_k]) := \langle v_0, \dots, v_k \rangle$. Obviously, $\langle v_0, \dots, v_k \rangle = \text{sgn}(\langle v_{s(0)}, \dots, v_{s(k)} \rangle)$ for any permutation s . In particular, $\langle v_0, \dots, v_k \rangle = 0$ if it has duplicate vertices. In fact, $C_k(K)$ is also a free abelian group. A basis is the set of all the oriented k -simplices with a selected orientation.

Proposition 7.1. For each k -simplex $\sigma = \{v_0, \dots, v_k\} \in K$, select an order of its vertices and write it sequentially inside small parentheses. If the selected order is $v_0 < v_1 < \dots < v_k$ then denote it as $(v_0, \dots, v_k)$. Let us call it the simplex with selected order associated with σ . Then $C_k(K) := S_k^{\text{ord}}(K)/R_k$ is isomorphic to a free abelian group F freely generated by the set of all k -simplex with selected order in K .

Proof. Let us consider the following diagram:

$$\begin{array}{ccc}
S_k^{\text{ord}}(K) & \begin{array}{c} \xleftarrow{\varphi_k} \\ \xrightarrow{\psi_k} \end{array} & F \\
\downarrow p & \nearrow \tilde{\varphi}_k & \\
C_k(K) := S_k^{\text{ord}}(K)/R_k & &
\end{array}$$

For each $[v_0, \dots, v_k] \in S_k^{\text{ord}}(K)$, if it has no duplicate vertices, there is always a permutation $\pi \in S_{k+1}$ such that $(v_{s(0)}, \dots, v_{s(k)})$ is the simplex with selected order associated with the simplex $\{v_0, \dots, v_k\}$. Define

$$\varphi_k([v_0, \dots, v_k]) = \begin{cases} 0 & \text{if has duplicate vertices} \\ \text{sgn}(\pi)(v_{s(0)}, \dots, v_{s(k)}) & \text{else} \end{cases}$$

and

$$\psi_k((v_0, \dots, v_k)) = [v_0, \dots, v_k]$$

p is the projection. Then $R_k \subseteq \ker \varphi_k$. One can verify it on each generator of R_k . Denote $p \circ \psi_k = \tilde{\psi}_k$. Verify $\tilde{\varphi}_k \circ \tilde{\psi}_k = \text{id}$ and $\tilde{\psi}_k \circ \tilde{\varphi}_k = \text{id}$ which are straightforward and the proposition is proved. $\square$

Definition 7.2. The oriented simplicial homology $H_k(K) := \ker \partial_k(K) / \text{im } \partial_{k+1}(K)$.

Remark 7.1. Just like singular chain group and singular homology, $C_\bullet : \text{SC} \rightarrow \text{cKom}$ is a functor from the category of simplicial complex to the category of chain complex; $H_\bullet : \text{SC} \rightarrow \text{Ab}$ is a functor from the category of simplicial complex to the category of abelian group. $\mu_\bullet : S_\bullet^{\text{ord}} \rightarrow C_\bullet$ is a natural transformation. It should be very straightforward to see this.

Remark 7.2. Likewise, we can also define relative simplicial homology $H_\bullet(K, L)$ and reduced simplicial homology $\tilde{H}(K)$. Mayer-Vietoris sequence also holds for simplicial homology.

Example 7.1. K is a simplicial complex and $s \in K$. Then the chain complex

$$\cdots \rightarrow C_k(\bar{s}) \rightarrow C_{k-1}(\bar{s}) \rightarrow \cdots \rightarrow C_1(\bar{s}) \rightarrow C_0(\bar{s}) \rightarrow \mathbb{Z} \rightarrow 0$$

has zero homology.

Proof. Let us construct a homomorphism $h_k : C_k(\bar{s}) \rightarrow C_{k+1}(\bar{s})$, $k \geq -1$ ($C_{-1}(\bar{s}) = \mathbb{Z}$). Recall that $C_k(\bar{s})$ is freely generated by k -dimensional oriented simplices $\{\langle v_0, \dots, v_k \rangle\}$ (or 1 when $k = -1$). If $k \geq n$, then $h_k = 0$. Otherwise, choose a vertex v in the vertex of $\bar{s}$ and let $h_k(\langle v_0, \dots, v_k \rangle) = \langle v, v_0, \dots, v_k \rangle$ (For the case $k = -1$, $h_{-1}(1) = \langle v \rangle$). It is straightforward to verify that $h_{k-1} \circ \partial_k + \partial_{k+1} \circ h_k = \text{id}$. Therefore, $\tilde{H}_k(\bar{s}) = 0$ $\square$

A natural question can be raised here: is the oriented simplicial homology isomorphic to the ordered simplicial homology? In fact, $\mu_\bullet$ is a chain map. To address the problem, we can prove that $\mu_\bullet$ is a chain equivalence. Once this question is answered, we can immediately conclude that $H_k(K)$ and $S_k(|K|)$ are isomorphic.

Theorem 7.1. $\mu_k : S_{\bullet}^{\text{ord}}(K) \rightarrow C_{\bullet}(K)$ is a chain equivalence for all simplicial complex K .

We will use Acyclic Model Theorem to prove it. Please make sure you fully understand Acyclic Model Theorem before reading the proof of this theorem.

Proof. For a simplicial complex K , let $\mathcal{C}(K)$ be the category of all its subcomplexes, i.e., $\text{Ob}(\mathcal{C}(K)) = \{L \mid L \subseteq K : \text{subcomplex}\}$. If $L_1, L_2 \in \text{Ob}(\mathcal{C}(K))$, then

$$\text{Hom}_{\mathcal{C}(K)}(L_1, L_2) = \begin{cases} L_1 \hookrightarrow L_2 & L_1 \subseteq L_2 \\ \emptyset & L_1 \not\subseteq L_2 \end{cases}$$

We now view S_k^{ord} and C_k as functors from $\mathcal{C}(K)$ to Ab .

1. Let $\mathfrak{M} = \{\bar{s} \mid s \in K\}$. For each $\bar{s} \in \mathfrak{M}$, $U_{\bar{s}} := \{[v_0, \dots, v_k] \mid \{v_0, \dots, v_k\} = s\} \subseteq S_k^{\text{ord}}(\bar{s})$. Let $U_k := \{U_{\bar{s}} \mid \bar{s} \in \mathfrak{M}\}$, then S_k^{ord} is U_k -expressible, $k \in \mathbb{Z}$.
2. Let $\mathfrak{M}_k := \{\bar{s} \mid s \in K, \dim s = k\}$. $\forall \bar{s} \in \mathfrak{M}_k$, $V_{\bar{s}} := \{\langle v_0, \dots, v_k \rangle \mid \{v_0, \dots, v_k\} = s, v_0 < \dots < v_k\} \subseteq C_k(\bar{s})$. Let $V_k := \{V_{\bar{s}} \mid \bar{s} \in \mathfrak{M}_k\}$, then C_k is V_k -expressible.

Let us consider the following diagram and apply Acyclic Model Theorem to construct $\hat{\mu}_{\bullet}$.

$$\begin{array}{ccccccccccc} \dots & \xrightarrow{\partial_{k+1}} & S_k^{\text{ord}} & \xrightarrow{\partial_k} & S_{k-1}^{\text{ord}} & \xrightarrow{\partial_{k-1}} & \dots & \xrightarrow{\partial_1} & S_0^{\text{ord}} & \xrightarrow{\varepsilon} & \mathbb{Z} & \longrightarrow & 0 \\ & & \downarrow \mu_k & & \downarrow \mu_{k-1} & & & & \downarrow \mu_0 & & \downarrow \cong & & \\ \dots & \xrightarrow{\partial_{k+1}} & C_k & \xrightarrow{\partial_k} & C_{k-1} & \xrightarrow{\partial_{k-1}} & \dots & \xrightarrow{\partial_1} & C_0 & \xrightarrow{\varepsilon} & \mathbb{Z} & \longrightarrow & 0 \\ & & \downarrow \hat{\mu}_k & & \downarrow \hat{\mu}_{k-1} & & & & \downarrow \hat{\mu}_0 & & \downarrow \cong & & \\ \dots & \xrightarrow{\partial_{k+1}} & S_k^{\text{ord}} & \xrightarrow{\partial_k} & S_{k-1}^{\text{ord}} & \xrightarrow{\partial_{k-1}} & \dots & \xrightarrow{\partial_1} & S_0^{\text{ord}} & \xrightarrow{\varepsilon} & \mathbb{Z} & \longrightarrow & 0 \end{array}$$

Let us construct $\hat{\mu}_0$.

$$\begin{array}{ccccc} C_0 & \xrightarrow{\varepsilon} & \mathbb{Z} & \longrightarrow & 0 \\ & & \downarrow \cong & & \downarrow 0 \\ S_0^{\text{ord}} & \xrightarrow{\varepsilon} & \mathbb{Z} & \longrightarrow & 0 \end{array}$$

Put any model $\bar{s} \in \mathfrak{M}_0$ into this diagram,

$$\begin{array}{ccccc} C_0(\bar{s}) & \xrightarrow{\varepsilon} & \mathbb{Z} & \longrightarrow & 0 \\ & & \downarrow \cong & & \downarrow 0 \\ S_0^{\text{ord}}(\bar{s}) & \xrightarrow{\varepsilon} & \mathbb{Z} & \longrightarrow & 0 \end{array}$$

by Example 7.1, the second row is exact, C_0 is V_0 -expressible. Then by Acyclic Model Theorem, there exists a natural transformation $\hat{\mu}_0 : C_0 \rightarrow S_0^{\text{ord}}$ such that the diagram commutes. For $k > 0$, the proof is similar by induction.

Next, let us briefly prove that $\mu_\bullet \circ \hat{\mu}_\bullet$ is chain homotopic to id and $\hat{\mu}_\bullet \circ \mu_\bullet$ is also chain homotopic to id. Consider the following diagram,

$$\begin{array}{ccccc}
S_0^{\text{ord}} & \xrightarrow{\varepsilon} & \mathbb{Z} & \longrightarrow & 0 \\
\hat{\mu}_0 \circ \mu_0 - 1 \downarrow & \swarrow 0 & \downarrow 0 & \swarrow 0 & \\
S_1^{\text{ord}} & \xrightarrow{\partial_1} & S_0^{\text{ord}} & \xrightarrow{\varepsilon} & \mathbb{Z}
\end{array}$$

Since S_0^{ord} is U_0 -expressible and the second row is exact for all models $\bar{s} \in \mathfrak{M}_0$ (Example 6.1), so by Acyclic Model Theorem there exists a natural transformation $h_0 : S_0^{\text{ord}} \rightarrow S_1^{\text{ord}}$ such that $\partial_1 \circ h_0 + 0 \circ \varepsilon = \hat{\mu}_0 \circ \mu_0 - 1$. For the cases where $k > 0$, the proof is similar. So by induction, $\hat{\mu}_\bullet \circ \mu_\bullet$ is chain homotopic to id. Similarly, $\mu_\bullet \circ \hat{\mu}_\bullet$ is also chain homotopic to id. Therefore, $S_\bullet^{\text{ord}}(K)$ is isomorphic to $C_\bullet(K)$. $\square$

Corollary 7.1. *Let K be a simplicial complex. Then $H_k(K) \xleftarrow{\mu_*} H_k^{\text{ord}}(K) \xrightarrow{\alpha_*} H_k(|K|)$, $k \in \mathbb{Z}$ are isomorphic. However, there is no direct construction of isomorphism between $H_k(|K|)$ and $H_k(K)$.*

Definition 7.3. *Let X be a topological space. The Euler characteristic number of X , $\chi(X) := \sum_{k \in \mathbb{Z}} (-1)^k \text{rank } H_k(X)$ if it is a finite sum.*

Corollary 7.2. *If X is a compact polyhedron, then $X \cong |K|$ for some finite simplicial complex K , so $\chi(X)$ is well-defined.*

Theorem 7.2. *$C_\bullet$ is a chain complex, then $\sum_{k \in \mathbb{Z}} (-1)^k \text{rank } H_k = \sum_{k \in \mathbb{Z}} (-1)^k \text{rank } C_k$ if the sum is finite and each term has a finite rank. The left term is called Euler Characteristics and the right term is called Euler form. From this theorem, we can see for any compact polyhedron $X \cong |K|$ where K is a finite simplicial complex, Euler characteristics depend on topology so the Euler form does not depend on triangulation of X .*

Proof. We have two exact sequences:

$$0 \rightarrow Z_k(C_\bullet) \rightarrow C_k \rightarrow B_{k-1}(C_\bullet) \rightarrow 0$$

and

$$0 \rightarrow B_k(C_\bullet) \rightarrow Z_k(C_\bullet) \rightarrow H_k(C_\bullet) \rightarrow 0$$

Since $C_\bullet$ is finitely generated module over PID ($\mathbb{Z}$), so their submodules $Z_\bullet(C_\bullet)$ and $B_\bullet(C_\bullet)$ have finite rank. So

$$\begin{aligned}
\text{rank } C_k &= \text{rank } Z_k(C_\bullet) + \text{rank } B_{k-1}(C_\bullet) \\
\text{rank } H_k(C_\bullet) &= \text{rank } Z_k(C_\bullet) - \text{rank } B_k(C_\bullet)
\end{aligned}$$

Therefore

$$\begin{aligned}
\sum_{k \in \mathbb{Z}} (-1)^k \text{rank } H_k(C_\bullet) &= \sum_{k \in \mathbb{Z}} (-1)^k \text{rank } Z_k(C_\bullet) - \sum_{k \in \mathbb{Z}} (-1)^k \text{rank } B_k(C_\bullet) \\
&= \sum_{k \in \mathbb{Z}} (-1)^k (\text{rank } Z_k(C_\bullet) + \text{rank } B_{k-1}(C_\bullet)) = \sum_{k \in \mathbb{Z}} (-1)^k \text{rank } C_k
\end{aligned}$$

$\square$

8 Simplicial Approximation

In this section we will introduce an important concept called simplicial approximation. That is, for a continuous map $f : |K_1| \rightarrow |K_2|$, we want to find a simplicial map $\varphi : V_{K_1} \rightarrow V_{K_2}$ whose realization $|\varphi|$ is homotopic to f . Before introducing the concept we will first introduce some concepts which is the continuation of chapter 3.1 of spanier's book or section 4 of this note.

Definition 8.1. K is a simplicial complex. $v \in V_k$ is a vertex. $\text{st } v := \{\alpha \in |K| \mid \alpha(v) \neq 0\}$.

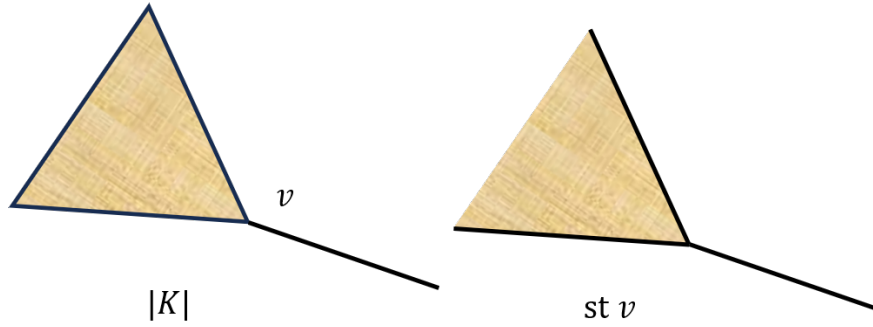


Figure 4: An example of $\text{st } v$

Proposition 8.1. $\text{st } v$ is open in $|K|$.

Proof. $\text{st } v$ is open $\Leftrightarrow (\text{st } v)^c = \{\alpha \in |K| \mid \alpha(v) = 0\}$ is closed $\Leftrightarrow \{\alpha \in |K| \mid \alpha(v) = 0\} \cap |s|$ is closed for each $s \in K$. Suppose $s = \{v_0, \dots, v_k\}$, so $\{\alpha \in |K| \mid \alpha(v) = 0\} \cap |s| = \{\sum_{i=0}^k t_i v_i \in |s| \mid t_j = 0 \text{ if } v_j = v\}$ which is either $|s|$ or the closed face expanded by $\{v_0, \dots, v_{j-1}, v_{j+1}, \dots, v_k\}$ and both are closed in $|s|$. Therefore, $\text{st } v$ is open in $|K|$. $\square$

Corollary 8.1. $\{\text{st } v \mid v \in V_k\}$ is an open cover of $|K|$.

Here is another description of $\text{st } v$:

Proposition 8.2. $\text{st } v = \cup_{s \in K} \langle s \rangle$

Proof. $\alpha \in \text{st } v \iff \alpha(v) \neq 0 \iff v \in \text{supp } \alpha \iff$. Therefore, $\text{st } v := \{\alpha \in |K| \mid \alpha(v) \neq 0\} = \{\alpha \in |K| \mid v \in \text{supp}(\alpha)\} = \cup_{s \in K} \langle s \rangle$. $\square$

Lemma 8.1. K is a simplicial complex. L is a subcomplex of K . $v_0, \dots, v_q \in V_K$. Then $\{v_0, \dots, v_q\} \in L \iff (\cap_{i=0}^q \text{st } v_i) \cap |L| = \emptyset$.

Proof. $\square$

Definition 8.2. K_1 and K_2 are two simplicial complexes and $f : |K_1| \rightarrow |K_2|$ is continuous. A simplicial map $\varphi : V_{K_1} \rightarrow V_{K_2}$ is a simplicial approximation of f if for every $\alpha \in |K_1|$, $f(\alpha) \in \langle s \rangle$, $s \in K_2$, then $|\varphi|(\alpha) \in |s|$.

Remark 8.1. *This definition may be weird but actually if φ is a simplicial approximation then φ is homotopic to f . This is because for any $\alpha \in |K_1|$, since $f(\alpha) \in \langle s \rangle$ and $|\varphi|(\alpha) \in |s|$, then let $h(\alpha, t) = (1 - t)f(\alpha) + |\varphi|(\alpha)$.*

Lemma 8.2. *K_1 and K_2 are two simplicial complexes and $f : |K_1| \rightarrow |K_2|$ is a continuous map. A map $\varphi : V_{K_1} \rightarrow V_{K_2}$ (not necessarily a simplicial map) is a simplicial approximation of $f \iff f(\text{st } v) \subseteq \text{st } \varphi(v), \forall v \in V_{K_1}$.*